

FABRICATION OF STRAIN WAVE HELICAL GEAR

A PROJECT-I REPORT [BMEE497J]

Submitted in partial fulfilment for the award of the degree of

Bachelor of Technology

in

**Mechanical Engineering [Automotive Engineering]
Mechanical Engineering**

by

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School of Mechanical Engineering



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Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

NOVEMBER 2024

DECLARATION BY THE CANDIDATES

We hereby declare that the project report entitled “**FABRICATION OF STRAIN WAVE HELICAL GEAR**” submitted by me to Vellore Institute of Technology, Vellore in partial fulfilment of the requirement for the award of the degree of **Bachelor of Technology in Mechanical Engineering** is a record of Bonafide work for Project-I [BMEE497J] carried out by us under the supervision of **Prof. Bikash Routh**. We declare that this report represents my concepts written in my ownwords and where others' ideas or words have been included, we have adequately cited and referenced the sources. We further declare that We have adhered to all principles of academic honesty and integrity and have not misrepresented fabricated or falsified any idea/data/fact/source in my submission. We understand that any violation of the above will cause disciplinary action by the Institute and can also invoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed. Further, we affirm that the contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

Place: Vellore

Signature of the Candidates

Date: 11/11/2024



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School of Mechanical Engineering

BONAFIDE CERTIFICATE

This is to certify that the Project-I [BMEE497J] report entitled “**FABRICATION OF STRAIN WAVE HELICAL GEAR**” submitted by **Tejeshwar S M (21BMA0013)** and **Balahariharan JJ (21BME0476)** to Vellore Institute of Technology, Vellore, in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in Mechanical Engineering** is a record of Bonafide work for the Project-I [BMEE497J] carried out by him/her under my guidance. The project fulfills the requirements as per the regulations of this institute and in my opinion meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

Project Guide

Head of the Department

Examiner (1)

Examiner (2)

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Place: Vellore

(Tejeshwar S M)

Date : 11/11/2024

(Balahariharan J J)

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EXECUTIVE SUMMARY

Strain wave helical gears, or harmonic drive gears, are precision mechanisms that provide high torque, compact size, and excellent positional accuracy and are ideal for robotics, aerospace, and space applications. Manufacturing these gears requires high precision and an excellent surface finish, significantly impacting their durability and quality. Our project aims to identify the optimal manufacturing process for a specific strain-wave helical gear for high-precision applications. By conducting comprehensive virtual analyses and simulations, we will evaluate various manufacturing processes, which will help us assess key factors, such as material properties, precision, surface finish, and production efficiency. The insights gained will guide the selection of the best manufacturing process, ensuring high-quality and reliable gear for advanced technological applications.

Keywords: *Strain wave helical gear, Manufacturing process, Computational Fluid Dynamics, HPDC*

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LIST OF SYMBOLS AND ABBREVIATIONS

psi	Pound per square inch
mm	Millimetre
kg	Kilogram
°C	Celsius
K	Kelvin
Pa	Pascal
C	Carbon
3D	Three Dimensional
HPDC	High-pressure Die Casting
LPDC	Low-pressure Die Casting
CFD	Computational Fluid Dynamics
CNC	Computer Numeric Control
FSI	Fluid-Structure Interaction
MPa	Mega Pascal

CHAPTER 1

INTRODUCTION AND LITERATURE REVIEW

1.1 Introduction

Strain wave helical gears, also known as harmonic drive gears, are a type of precision gear mechanism that provides high torque, compact size, and excellent positional accuracy. They are commonly used in applications that require precise motion control, such as robotics, aerospace, and space exploration.

1.2 Harmonic Drive

Harmonic drive also known as Strain wave gearing is a type of mechanical gear system known for providing high precision, compact size, and large gear reduction ratios, commonly used in robotics, aerospace, and precision machinery.

The system works based on the principles of elastic deformation. It consists of three main parts: the circular spline, the flex spline, and the wave generator.

Wave Generator: The wave generator is a circular cam with an elliptical outline, typically surrounded by bearings. When it rotates, it forces the flexible part of the gear system (the flex spline) into an elliptical shape. The motor or input shaft is usually connected to the wave generator, which rotates it.

Flex spline: The flex spline is a thin, flexible, cup-shaped component with external teeth around its open end. It is made from a material that allows it to bend or deform elastically. The wave generator's rotation causes the flex spline to deform into an elliptical shape. This deformation allows certain teeth of the flex spline to engage with those of the circular spline at two points, positioned at opposite sides of the ellipse.

Circular Spline: A stiff ring with internal teeth and a marginally higher tooth count than a flex spline is called a circular spline. In many applications, it acts as a "fixed" component of the system, remaining stationary.

Working Mechanism:

As the wave generator rotates, it forces the flex spline to deform continuously, causing its teeth to engage with the circular spline in a rotating pattern. Since the flex spline has fewer teeth than the circular spline, each rotation of the wave generator advances the flex spline by only a small difference relative to the circular spline. This difference in movement leads to a significant gear reduction.

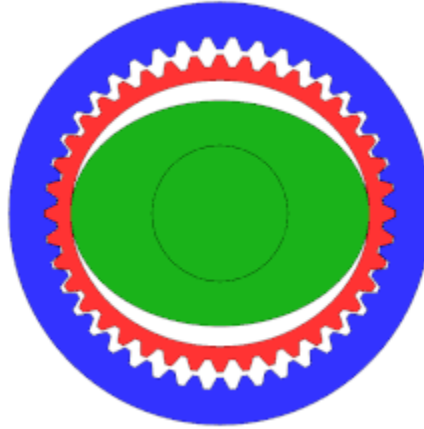


Figure 1.1: Harmonic Drive

Citation: Wikipedia contributors. (2024, February 18). *Strain wave gearing*. Wikipedia.

https://en.wikipedia.org/wiki/Strain_wave_gearing

1.3 Manufacturing Aspect

The manufacturing of strain-wave helical gears involves achieving high precision and excellent surface finish, which are critical for their performance. The selection of the manufacturing process affects the gear's durability, efficiency, and overall quality. Processes such as CNC machining, pressure die casting, and additive manufacturing have their advantages and limitations, necessitating a thorough evaluation to determine the best method for producing high-quality strain-wave helical gears.

In our project, we will conduct thorough research to select a suitable manufacturing process based on various factors such as material availability, cost, and production time. After selecting a suitable process, we will conduct various virtual simulations for manufacturing our gear using the selected process and analyse the effect of multiple parameters such as temperature and pressure.

1.3.1 High-Pressure Die Casting

A manufacturing technique called high-pressure die casting (HPDC) is used to create intricate metal components with excellent accuracy and a smooth surface. Molten metal is injected under high pressure—typically between 1500 and more than 2500 psi—into a steel mould or die. High production rates and the construction of complex shapes that are challenging to accomplish with other casting techniques are made possible by the metal's rapid solidification within the mould.

1.3.2 Choosing HPDC over LPDC.

For the production of metal components, die-casting techniques offer a great lot of versatility. Parts made from nearly any metal in the desired shape can be produced by both HPDC and LPDC. On the other hand, non-ferrous metals including magnesium, aluminium, and zinc are processed using HPDCs. Parts made of aluminium alloys by high-pressure die-casting can even be as thin as 0.5 mm. Increased precision and consistency can lead to fewer defects and waste, which eventually lowers costs. However, high-pressure die casting is more costly than classical casting due to the increased cost of equipment and tools. Recently, the technique has been expanded to manufacture castings up to 3 kg in weight in stainless steel, copper and copper alloys, cobalt, carbon (up to 0.5%C), and air-melting grades of nickel alloys. We are therefore inspired to use the HPDC approach to work on the strain-wave helical gear.

1.3.3 Electro-Chemical Machining

Electrochemical Machining (ECM) is a non-conventional machining method that uses electrolysis instead of physical forces to remove metal. In this process, the tool serves as the cathode (Cu or Cu alloy such as Brass) and the workpiece as the anode (Al gear), with an electric current [Current density: 10-300 A/cm² and Voltage: 5-25 volts] flowing through an electrolyte solution (Sodium Nitrate (NaNO₃) or Sodium Chloride (NaCl) solution). In the gap between them, causing the metal on the workpiece to dissolve. ECM is capable of precisely machining hard materials and complex geometries beyond the capabilities of traditional machining methods. It results in a smooth surface finish, with no thermal or mechanical impact on the workpiece.

1.4 Literature Review

In recent studies on casting processes, the utilization of salt cores has emerged as a promising technique for achieving high dimensional stability and facilitating the removal of internal structures post-casting. Salt cores provide an effective solution for manufacturing complex internal geometries, especially in components like helical gears. By reducing internal defects and enhancing dimensional accuracy, salt cores present a viable approach for producing strain wave helical gears with intricate internal profiles.

Addressing the dimensional challenges inherent in casting processes, research has identified common defects – such as shrinkage, warping, and dimensional inaccuracies – that can undermine the performance of high-precision components like strain-wave helical gears. Effective control over casting parameters, along with the adoption of advanced simulation techniques, is crucial to predict and mitigate these defects, ensuring the production of high-quality gears.

Casting parameter optimization, including temperature control, cooling rates, and the choice of mould materials, has also been shown to significantly impact the final properties of cast components. Uniform microstructure and minimized residual stresses are critical for strain wave helical gears, as they directly influence durability and performance. Studies reveal that optimized casting parameters can enhance key mechanical properties, including hardness and tensile strength, making the gears more suitable for demanding applications.

Mould design plays a fundamental role in high-pressure die casting (HPDC), especially when considering stress components such as normal and shear stresses. Understanding stress distribution, including principal stresses and von Mises stress, has proven essential for designing moulds that ensure robust component quality and durability. The research underscores the value of stress analysis in predicting material behaviour and supporting effective mould design.

Temperature gradients during the casting process can induce thermal stresses that may cause deformation or even failure in the die and cast part. Recent findings highlight the importance of incorporating thermal expansion coefficients and specific heat capacities.

into simulation models to accurately predict these stresses. This approach is critical for maintaining the structural integrity of components, such as strain wave helical gears, which require exceptional reliability.

The use of strain gauges in conjunction with simulation techniques has further advanced the field by providing precise stress and strain measurements during the casting process. Comparisons between simulation outputs and strain gauge measurements validate model accuracy, offering a framework for converting strain data into actionable stress insights. Equations for principal strains, stresses, and von Mises stress have been instrumental in refining these analyses.

High-pressure die casting (HPDC) is frequently used for manufacturing light metal alloys like aluminium and magnesium due to its efficiency and the dimensional accuracy it imparts to castings. However, HPDC is prone to defects, such as entrapped air and shrinkage porosity, which can create microstructural inhomogeneity and impact mechanical properties, occasionally resulting in early material failure. Optimizing HPDC process parameters has therefore become essential to minimize defects and improve the consistency of mechanical properties.

Recent advancements in numerical modelling present an efficient alternative to traditional trial-and-error methods for HPDC optimization. The development of finite element models for HPDC, particularly for optimizing the piston slow shot profile, has shown promising results. Studies on A356 alloy demonstrate that modelling melt flow, heat transfer, solidification, and defect formation can enhance mechanical properties through parameter refinement. Further, accurate input of temperature-dependent material properties, such as yield stress and hardening coefficients, has proven essential in achieving reliable predictions of stress under operational conditions. Integrating elastic-plastic behaviour into simulation models ensures that strain wave helical gears can reliably perform in their intended applications, highlighting the importance of advanced modelling in casting technology.

1.5 Knowledge gained from literature

- In the HPDC process, oxides can be harmful because they can cause defects in the final casting process. A more uniform distribution of oxides indicates that these defects are less concentrated in specific areas, leading to better overall mechanical properties in the final product.
- Entrapped air can create voids in the metal, weakening the casting, whereas oxides can act as stress concentrators, both of which negatively impact the tensile properties. By optimizing the shot velocity, these issues can be minimized, resulting in stronger and more ductile castings.
- Casting simulation software provides an accurate representation of the stress in the die, and the simulation model will have better accuracy on the casting surface of the die, which will experience greater stress fluctuation during the casting cycle.

1.6 Gaps Identified

The above papers that we referred to used software such as Altair Inspire Cast to conduct various simulations, even for checking the strength of the die. We will use ANSYS Workbench (Fluent), which is an application of CFD. Using the CFD approach, we can assess the strength of the die owing to the application of high pressure and temperature at various points in the mould. We will also use the Altair Inspire Cast software to make many observations such as airflow, mould erosion, solid fraction, filling time, solidification time, porosity, and mould temperature.

1.7 Problem definition

The challenge lies in identifying the most suitable manufacturing method that can consistently produce strain wave helical gears meeting the desired specifications. Current literature lacks a comprehensive analysis comparing different manufacturing techniques such as additive manufacturing, precision casting, and CNC machining etc., specifically for strain wave helical gears. This lack of knowledge leads to inefficiencies, higher costs and lower gear performance.

1.8 Objectives of this project

- To perform a comprehensive cost-benefit analysis comparing the HPDC method to conventional manufacturing techniques for strain-wave helical gears, considering factors such as production speed, material usage, and overall manufacturing cost.
- To identify and evaluate alloys suitable for the HPDC process that meet the performance requirements of strain-wave helical gears, focusing on factors such as strength, wear resistance, and thermal stability.
- To develop and optimize die designs specifically tailored for the production of strain wave helical gears, ensuring accurate replication of complex gear geometries and minimizing defects such as porosity and warping.
- To conduct various virtual simulations of the flow of molten metal into a mould that will occur during HPDC.
- Virtual analyses will be conducted to test the strength of the die-cast mould under high-pressure and high-temperature conditions.

1.9 Motivation

One of the main problems that manufacturers face is fatigue failure or the inability to achieve a fine and smooth surface finish because the manufacturing process is conventional machining or sand casting. To solve this problem, we first die-cast the product using the high-pressure die-casting method, during which we design the mould such that we can achieve the smallest possible thickness of the gear. To further achieve the desired thickness, we machined it using the ECM method.

CHAPTER 2

METHODOLOGY AND EXPERIMENTAL WORK

2.1 PROJECT EXECUTION STAGES

Step 1: Analyze the relevant previous designs and read research papers related to the strain wave Helical Gear and HPDC.

Step 2: Creating CAD models for the required mould designs in SOLIDWORKS

Step 3: Simulating the constructed 3D model in Altair Inspire Cast to define and run the casting analysis

Step 4: Conducting CFD research on the design using Ansys Workbench

2.2 TECHNICAL SPECIFICATIONS

2.2.1 Conventional Harmonic Drive Gear

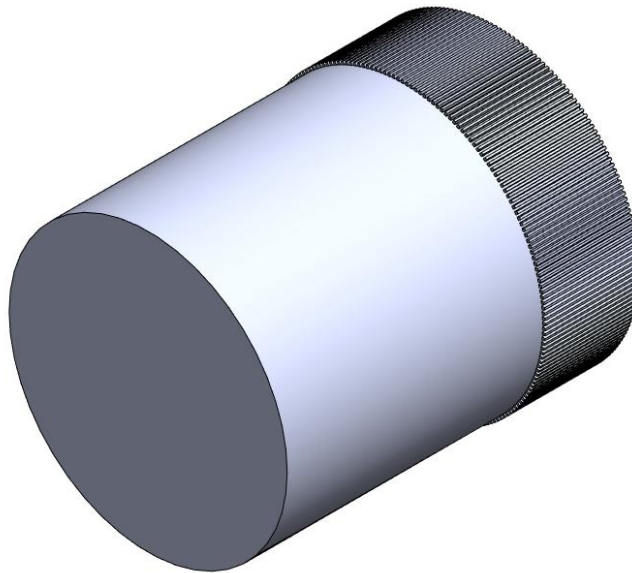


Figure 2.1

2.2.2 Conceptual Strain Wave Helical Gear

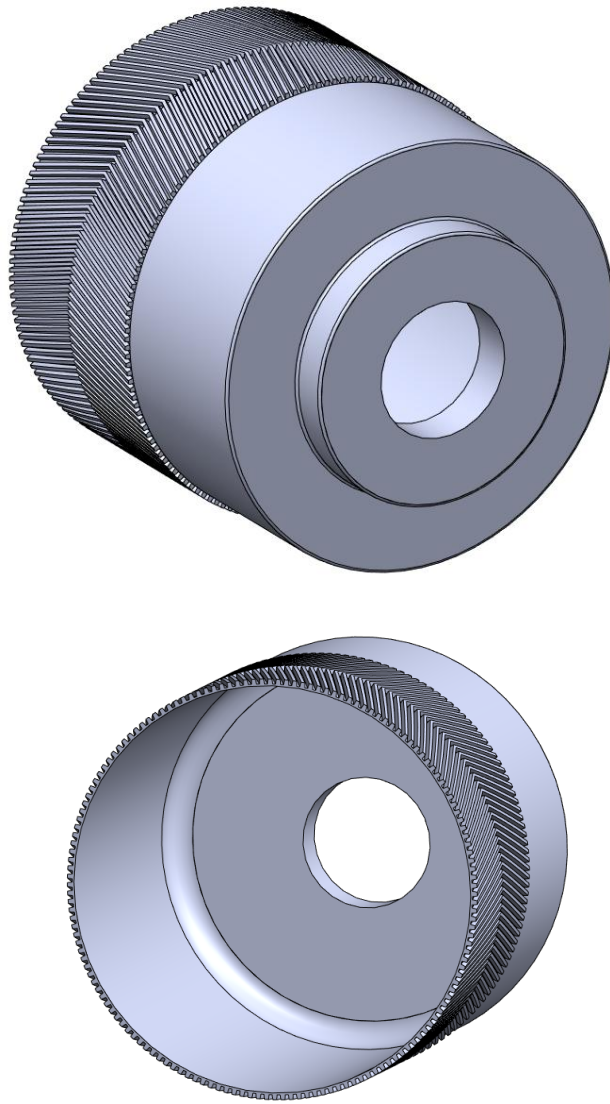


Figure 2.2

Gear specifications:

- **Radial Thickness: 1.4mm**
- **Diameter: 81.75mm**
- **Length: 100.05mm**
- **Thickness: 19.06mm**

2.2.3 Helical Gear Teeth Profile

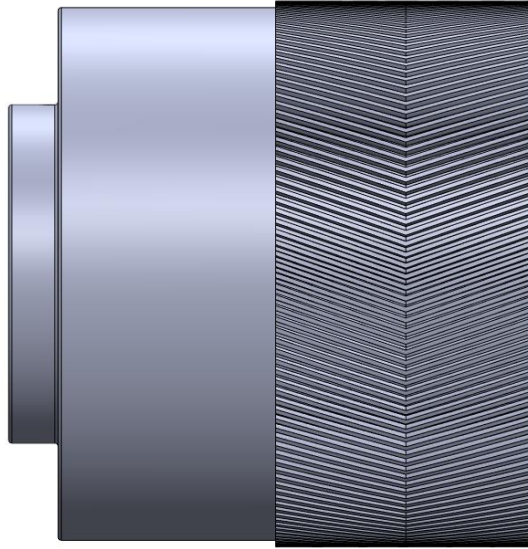


Figure 2.3

Teeth Specifications:

- Number of Teeth = 156
- Module = 0.5 mm
- Pitch circle diameter = 83.83 mm
- Addendum circle diameter = 84.59 mm
- Dedendum circle diameter = 82.56 mm
- Helix angle = 15°
- Tooth thickness = 21 mm
- Fillet radius = 0.05 mm

2.3 Strain Wave Helical Gear Mold

We discovered that this gear can be manufactured using two methods- back-stroke and front-stroke. So, we have designed moulds for the two methods to find which gives a better result in various aspects like stress, strain, deformation etc.

2.3.1 Back stroke:

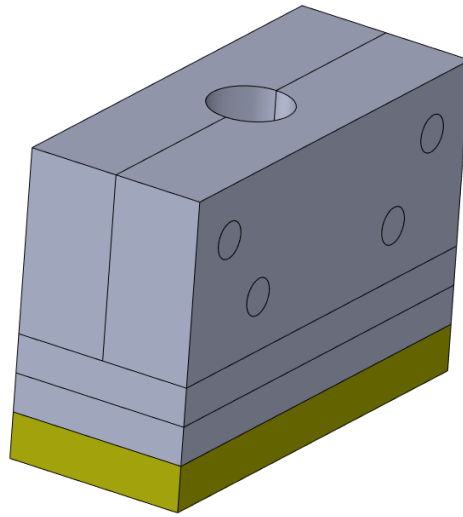
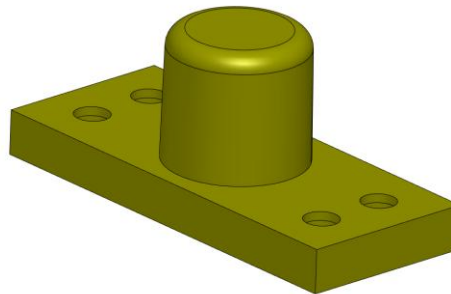


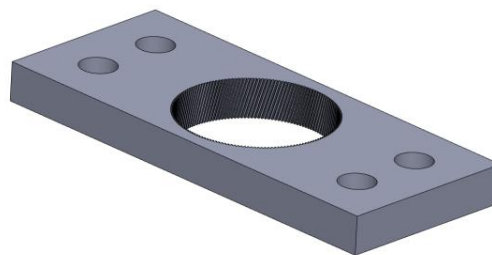
Figure 2.4: Back-stroke Mould

Mold components:

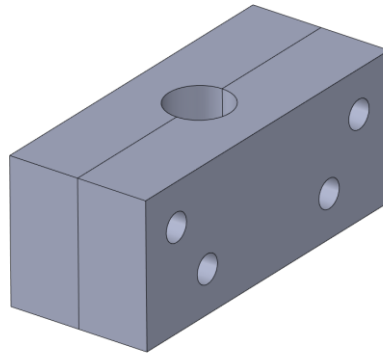
a) Core plate:



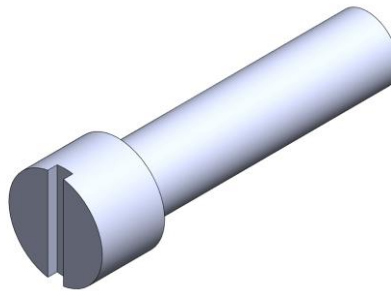
b) Teeth plate:



c) Top base:



d) Fastener (M20 cylinder head bolt):



Mould Major Dimensions:

- Length = 240 mm
- Width = 100 mm
- Height = 142 mm

2.3.2 Front stroke:

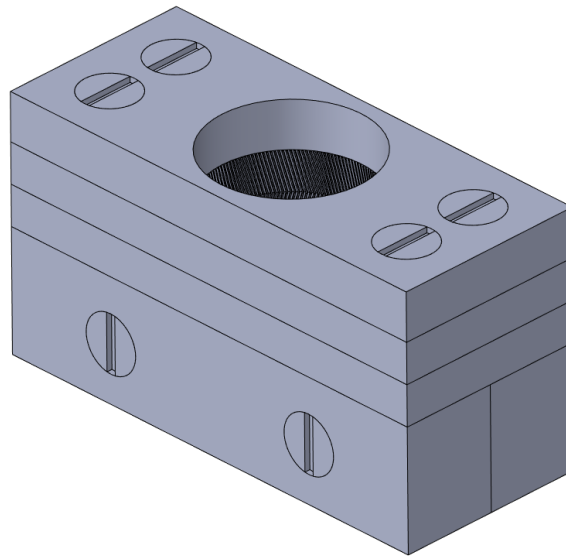
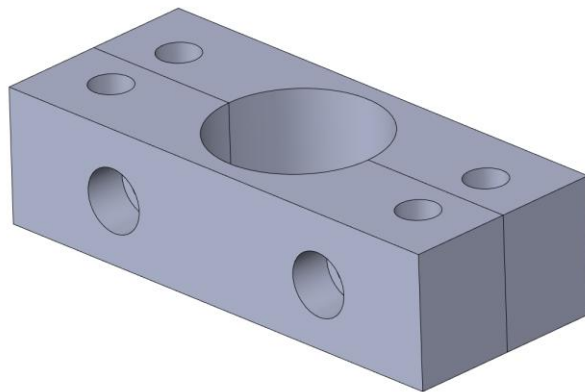


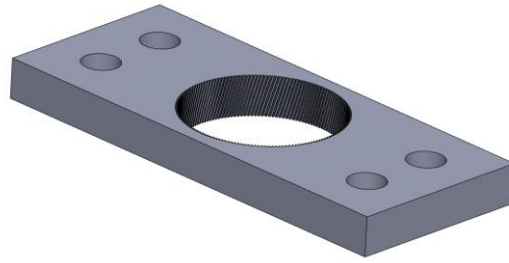
Figure 2.5: Front-stroke Mould

Mold components:

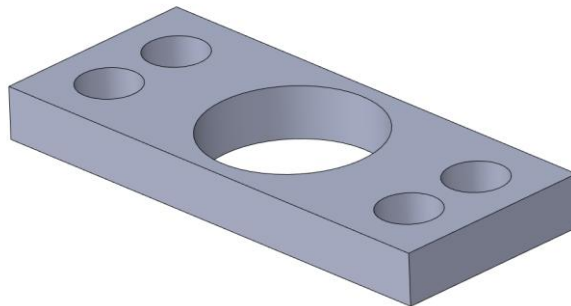
a) Bottom base:



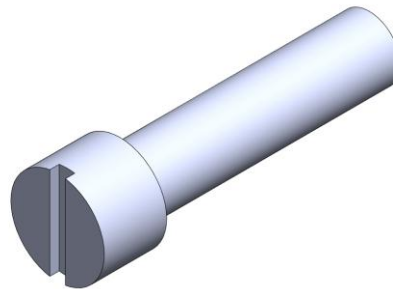
b) Teeth plate:



c) Top base:



d) Fastener (M20 cylinder head bolt):



Mold Specifications:

- Length = 240 mm
- Width = 100 mm
- Height = 130.4 mm

2.4 Die-Casting Parameters

- Pouring temperature = 660°C
- Die casting pressure = 80 mPa
- Mold temperature = 220°C
- Injection speed = 0.5 m/s to 2.5 m/s

2.5 Liquid Metal Properties (Aluminium):

- Density: 2357 kg/m³
- C_p (specific heat) = 1080 J/kgK
- Thermal conductivity = 97 W/mK
- Viscosity = 0.0012 kg/ms

2.6 Mould material - 1010 carbon steel annealed (Specifications) :

- Density: 7849.8 kg/m³
- C_p (specific heat) = 422.63 J/kgK
- Thermal conductivity: 56.592 W/mK
- Thermal Expansion coefficient: 1.14e-05 K⁻¹
- Young's Modulus: 2.12e+11 Pa
- Poisson's ratio: 0.29

2.6 Inspire Cast results

The use of the software called “**Altair Inspire Cast**” helps us to get a visual understanding of the HPDC process and using the results we can observe various parameters involved like temperature and pressure. Later on, we will be designing the mould that will be required to manufacture this gear and conducting various virtual tests on the mould to test its integrity.

2.6.1 Conventional Strain Wave Gear (Straight Teeth):

Back Stroke:

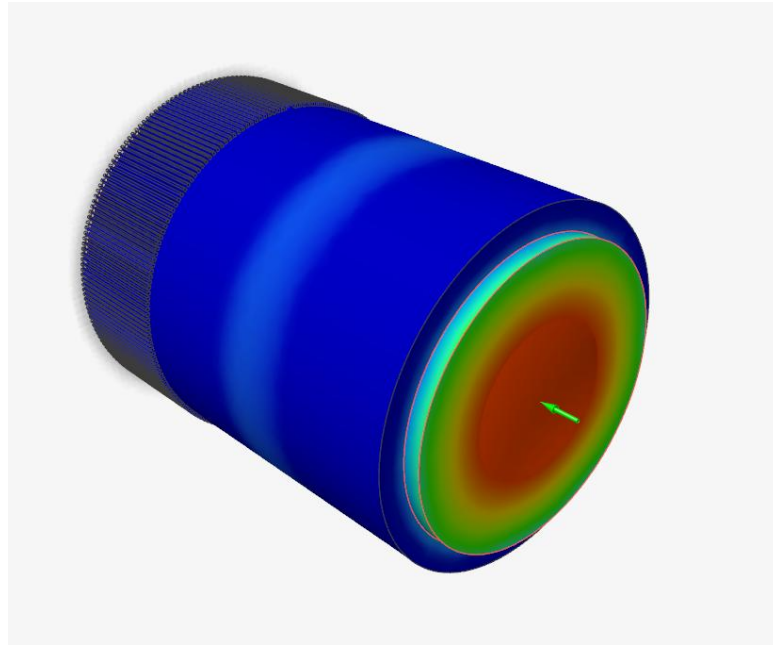


Figure 2.6: Pressure variation during filling

- **High-Pressure Areas (Red):** These regions indicate areas where the material is entering the mould quickly or where it is being compressed as it moves. If these pressures are too high, it might lead to defects such as porosity or incomplete filling.
- **Low-Pressure Areas (Blue):** These regions are where the material might not have reached yet or where it is moving more slowly. Extremely low pressures can also lead to defects like air entrapment.

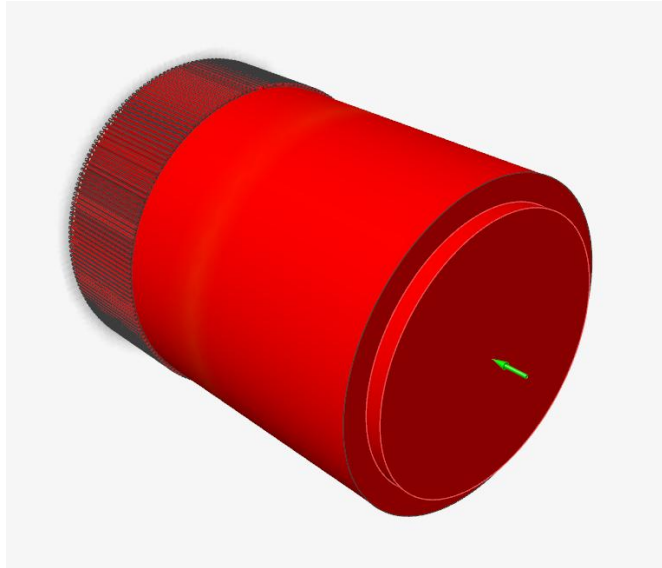


Figure 2.7: Temperature variation during filling

- **High-Temperature Areas (Red):** These are the areas present near the regions where the molten material enters the mould. High temperatures could cause defects like hot tears or cracking.
- **Low-Temperature Areas (Blue):** These are the regions that are usually present at the end of the fill path or in thin sections of the mould. Low-temperature areas can lead to premature solidification, which might result in incomplete filling (cold shuts) or misruns.

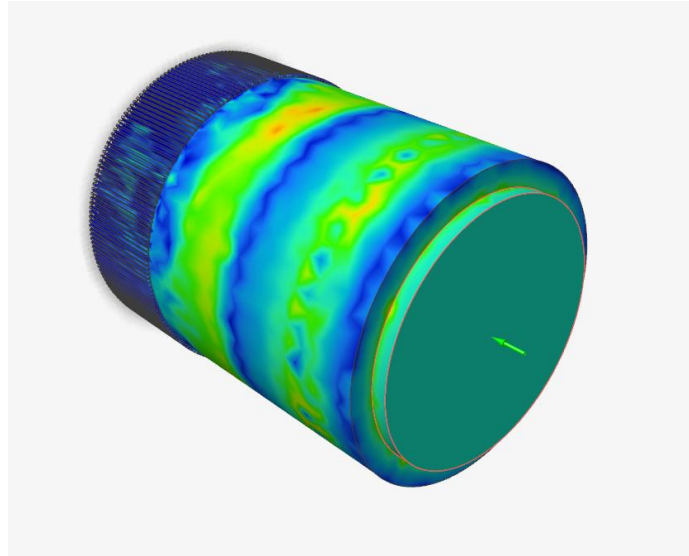


Figure 2.8: Mold erosion during filling

- Red areas indicate regions with the highest erosion velocity, which is where the mould is most likely to experience significant wear and tear.
- Blue areas represent low or minimal erosion, indicating regions of the mould that will endure the least damage.

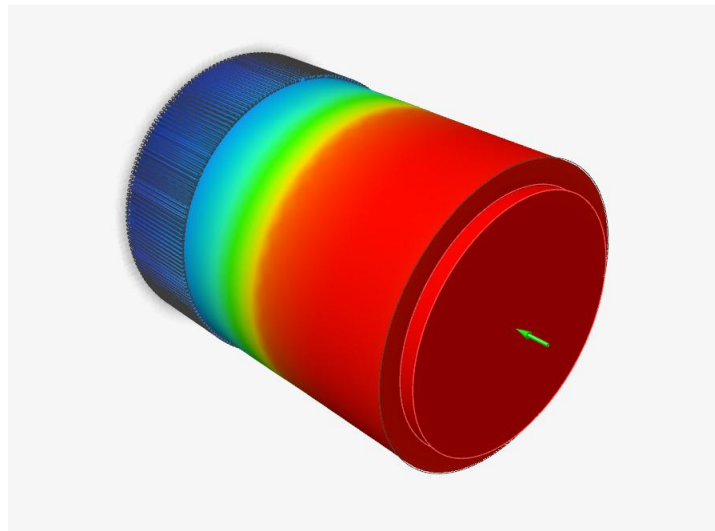


Figure 2.9: Temperature while solidifying

2.6.2 Conceptual Strain Wave Helical Gear:

Back stroke:

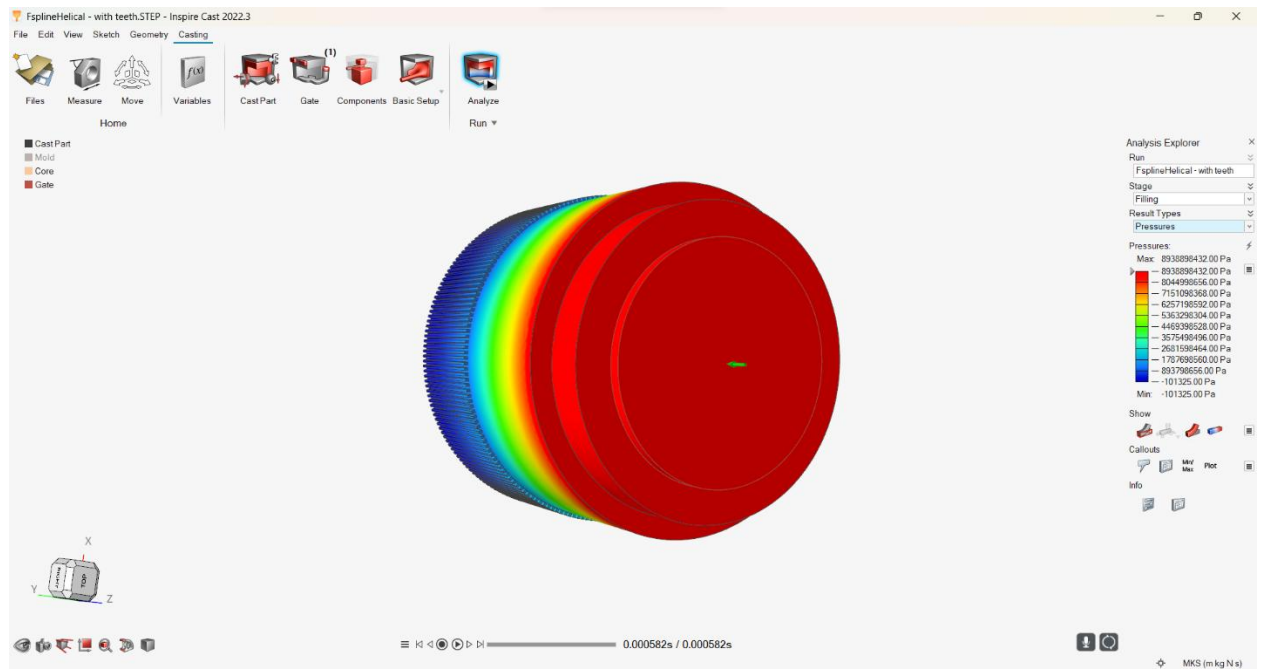


Figure 2.10: Pressure variation during filling

- **High-Pressure Areas (Red):** These regions indicate areas where the material is entering the mould quickly or where it is being compressed as it moves. If these pressures are too high, it might lead to defects such as porosity or incomplete filling.
- **Low-Pressure Areas (Blue):** These regions are where the material might not have reached yet or where it is moving more slowly. Extremely low pressures can also lead to defects like air entrapment.

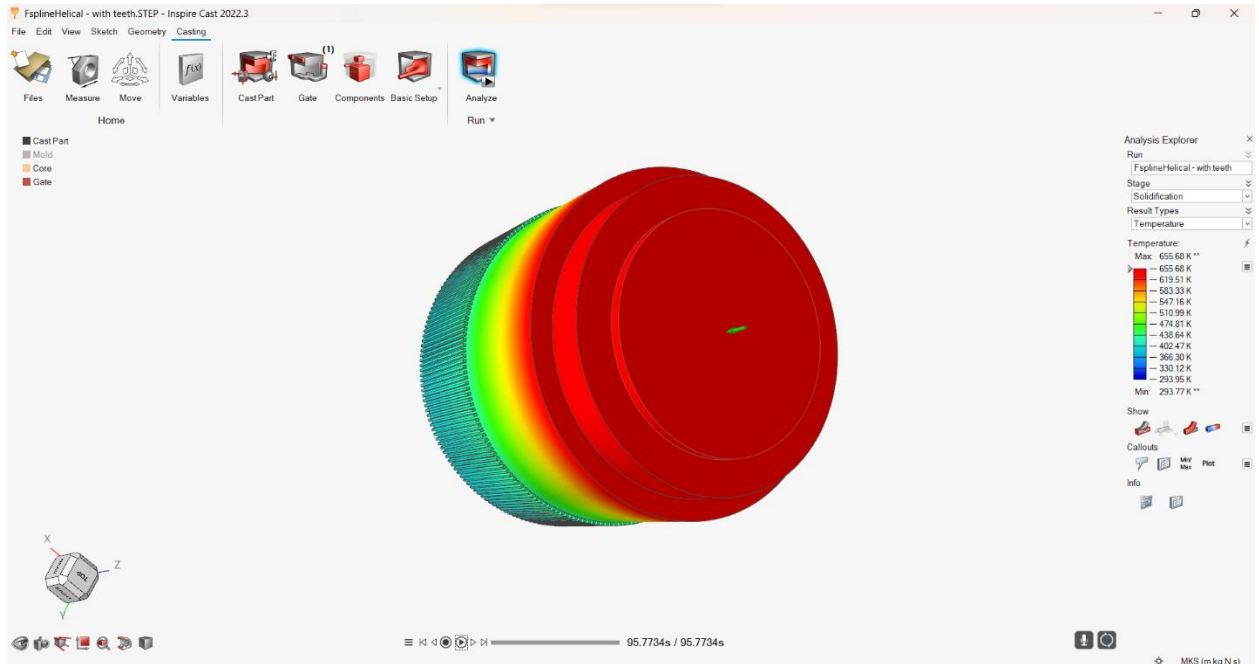


Figure 2.11: Temperature while solidifying

- **High-Temperature Areas (Red):** These areas are still hot and have not fully solidified yet. High temperatures may indicate slower cooling, leading to issues like shrinkage porosity or hot spots, where defects might occur due to uneven cooling rates.
- **Low-Temperature Areas (Blue/Green):** These regions have cooled down and likely solidified. Uniform cooling is generally desirable to minimize internal stresses and prevent defects.

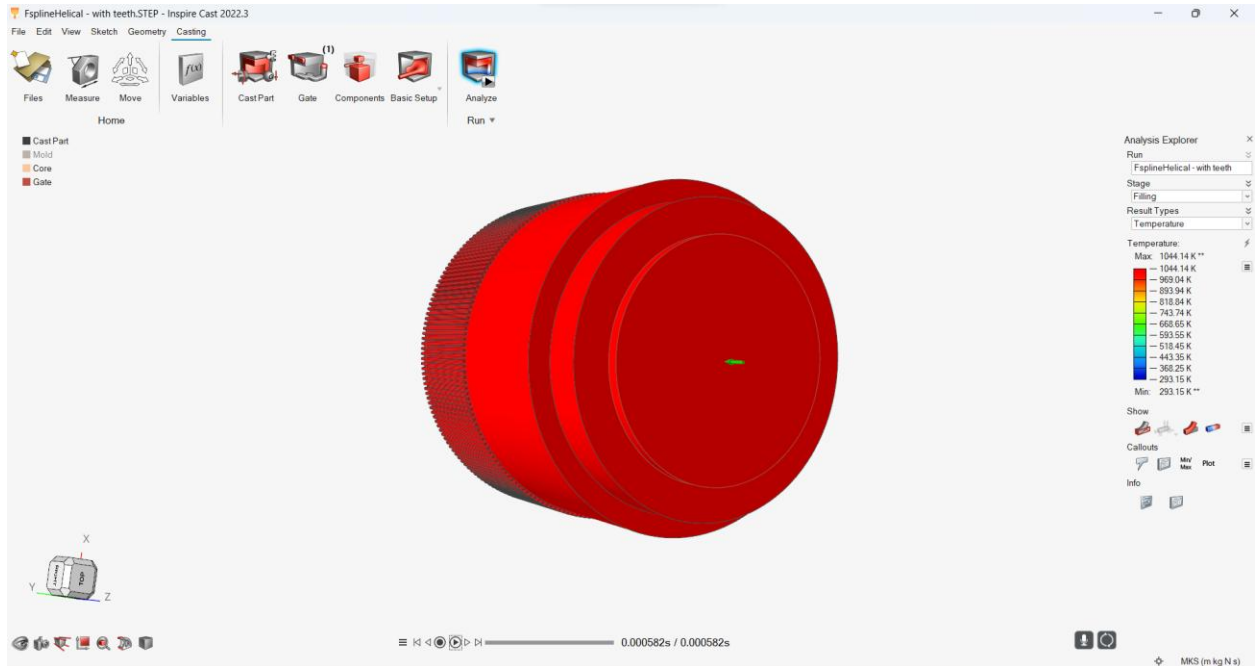


Figure 2.12: Temperature variation during filling

- **High-Temperature Areas (Red):** These are the areas present near the regions where the molten material enters the mould. High temperatures could cause defects like hot tears or cracking.
- **Low-Temperature Areas (Blue):** These are the regions that are usually present at the end of the fill path or in thin sections of the mould. Low-temperature areas can lead to premature solidification, which might result in incomplete filling (cold shuts) or misruns.

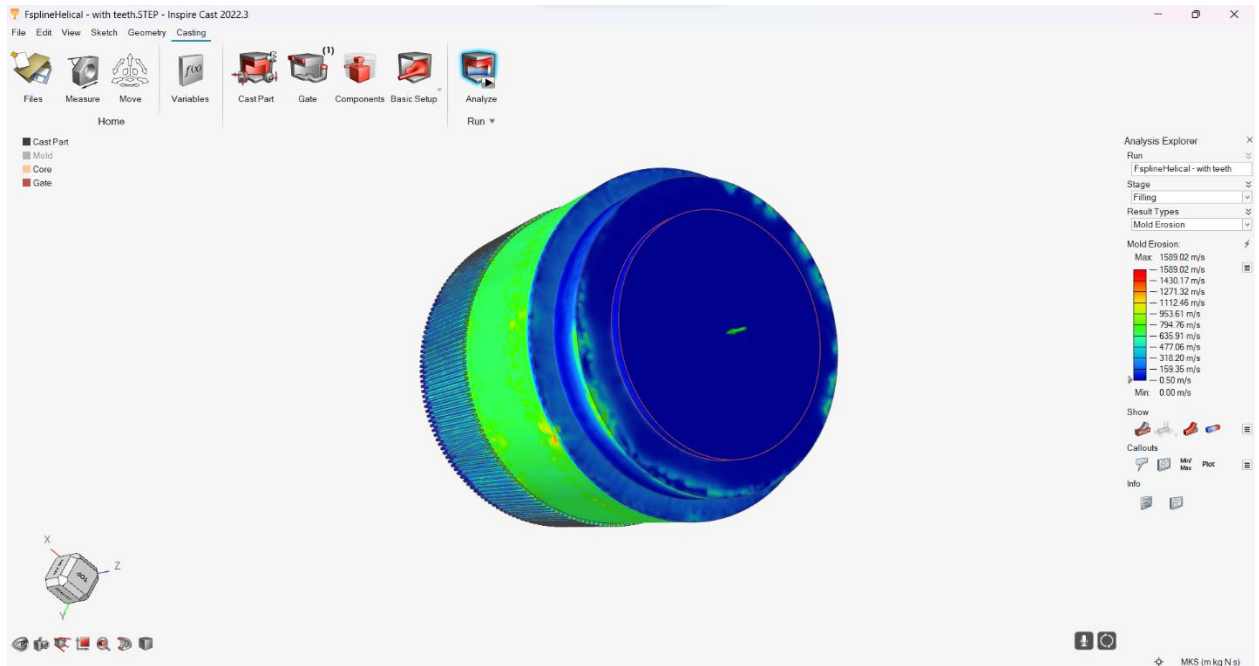


Figure 2.13: Mold erosion during filling

- Red areas indicate regions with the highest erosion velocity, which is where the mould is most likely to experience significant wear and tear.
- Blue areas represent low or minimal erosion, indicating regions of the mould that will endure the least damage.

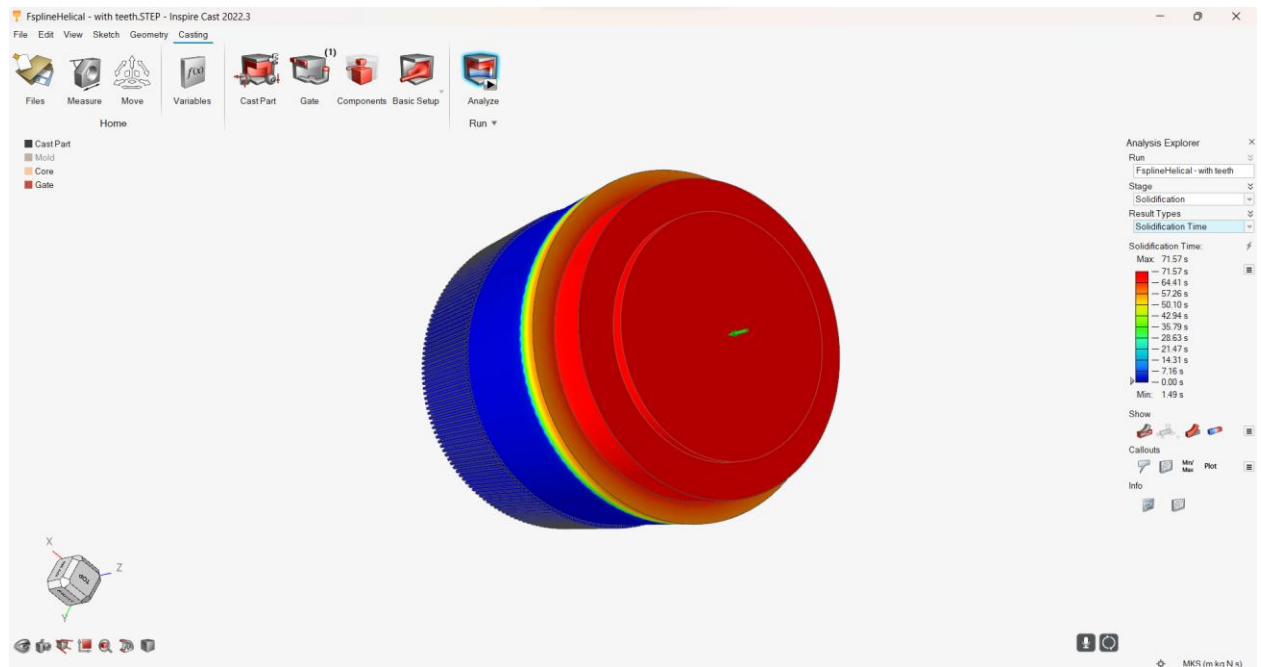


Figure 2.14: Variation of solidification time between the component

- High-Solidification time (Red): These are the areas that take much time to solidify. Here, a higher risk of shrinkage defects and porosity forming can be observed. Slower cooling rates can reduce thermal gradients, leading to lower residual stresses.
- Low-Solidification time (Blue): These are the areas that take little time to solidify. Lesser solidification time can reduce the risk of shrinkage defects as the metal solidifies more quickly, potentially before significant contraction occurs. Rapid cooling increases thermal gradients, which can induce higher residual stresses.

Front stroke:

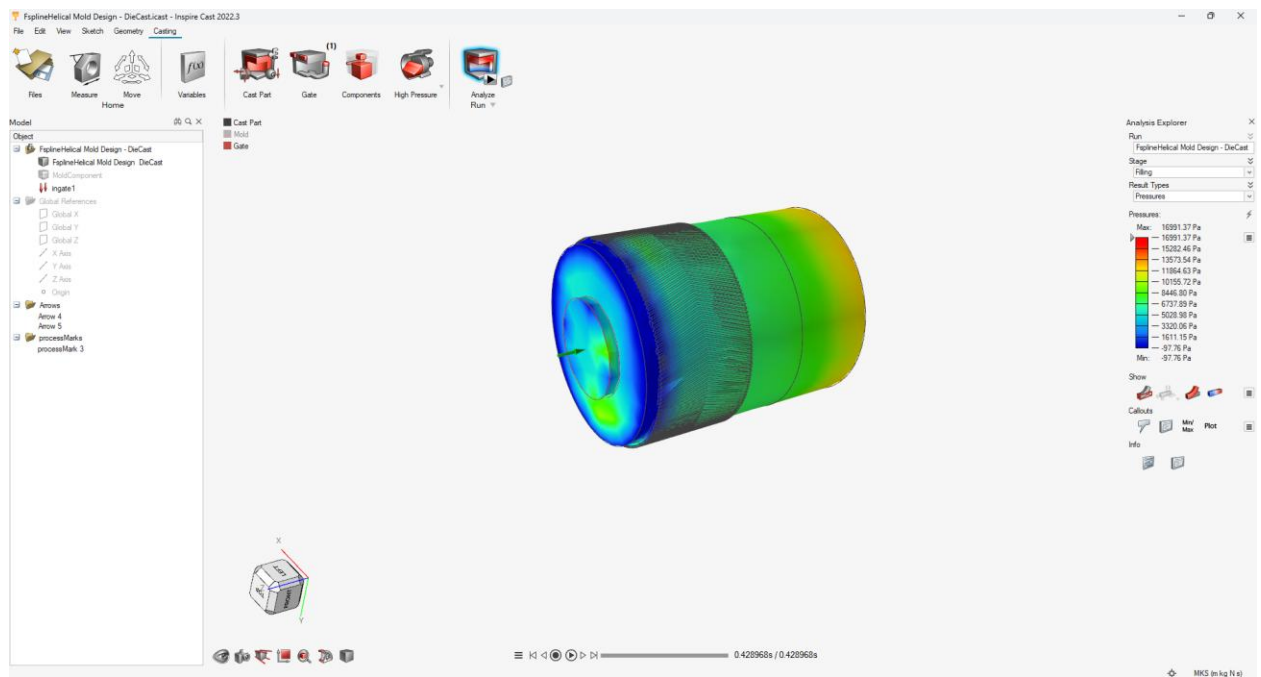


Figure 2.15: Pressure variation during filling

- **High-pressure areas (Red):** The red regions indicate where the molten metal experiences the most resistance or backpressure during filling. This can occur at bottlenecks, sharp corners, or areas furthest from the inlet. Identifying high-pressure areas helps to understand if there is any uneven flow or risk of defects like cold shuts (where the metal cools before filling the mould).
- **Low-Pressure areas (Blue):** The blue areas are regions where the molten metal flows more freely, facing less resistance, which could lead to potential filling defects like air entrapment, potential porosity or incomplete filling if not controlled properly.

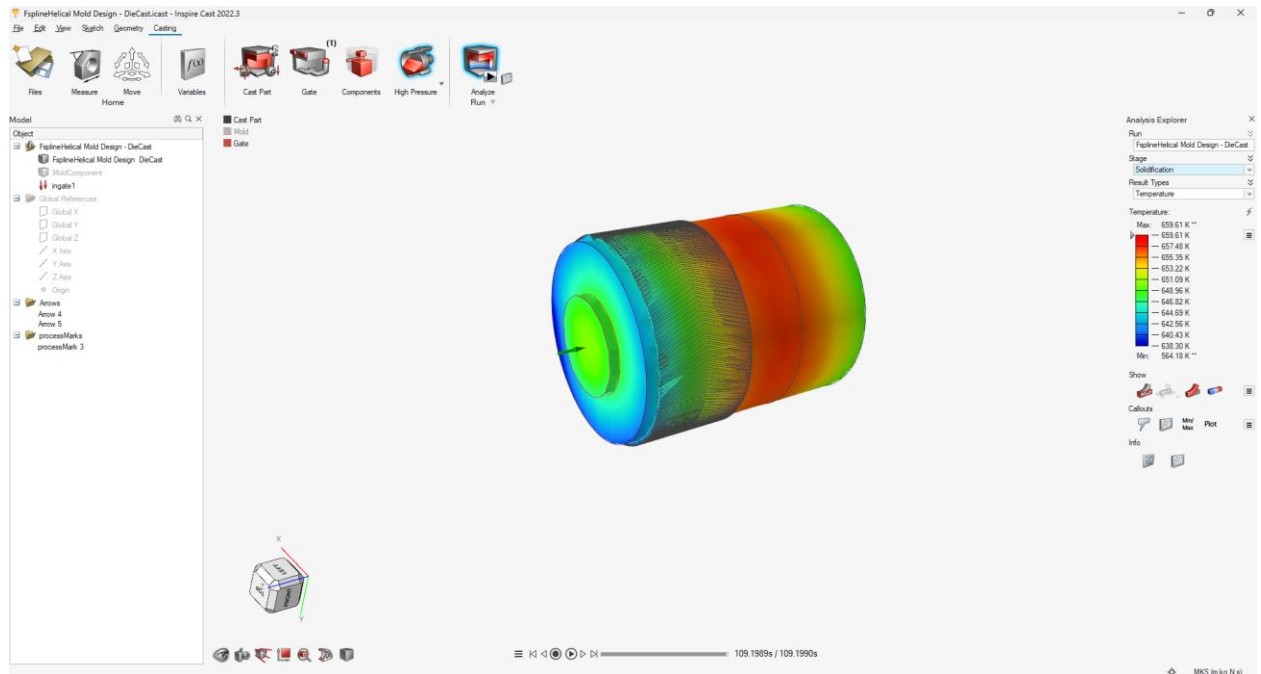


Figure 2.16: Temperature while solidifying

- **Hot Zones (Red/Orange):** These are areas where the part retains more heat and solidifies last. These zones can lead to defects like shrinkage porosity if cooling is uneven.
- **Cooler Zones (Blue/Green):** These regions have already solidified or are in the process of solidifying. Proper solidification in these zones ensures that the part forms as intended.
- If certain regions solidify too quickly or too slowly, it can lead to casting defects like hot tears, shrinkage, or warping. The uniformity of cooling is crucial for ensuring the structural integrity of the cast part.

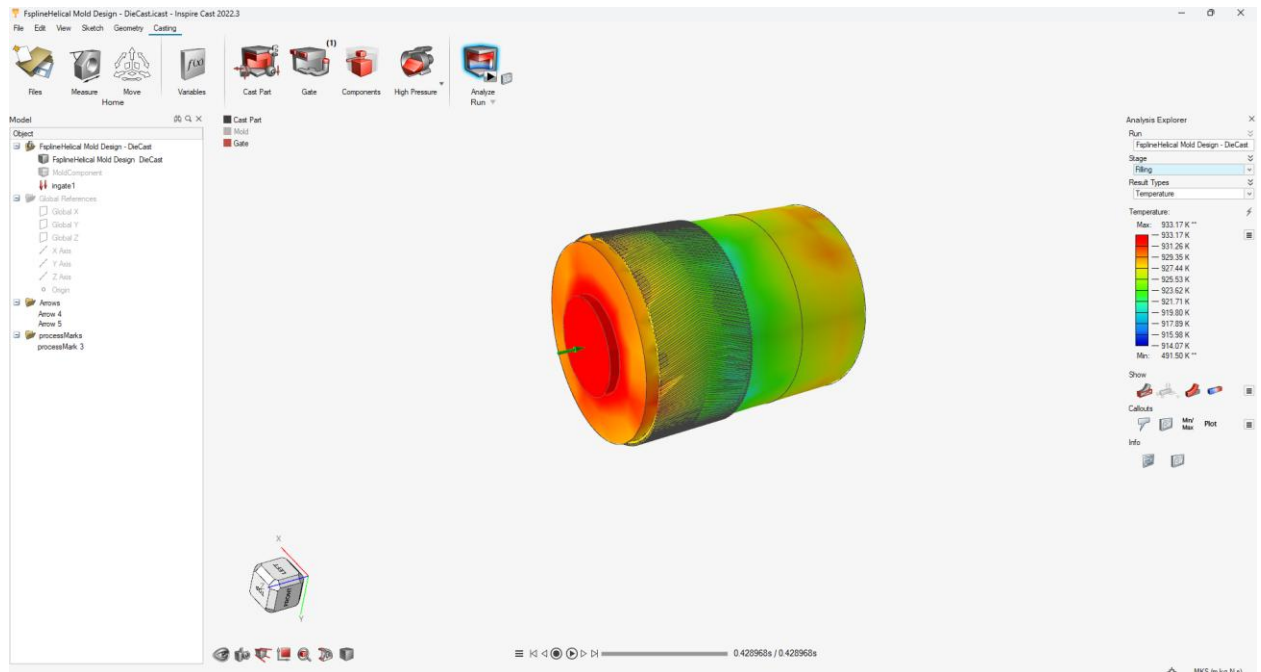


Figure 2.17: Temperature variation during filling

- **Hot Zones (Red/Orange):** These are the areas where the molten material first enters the mould and remains at a higher temperature. This happens because the material retains more heat as it flows.
- **Cooler Zones (Green/Yellow):** These areas represent regions where the molten material is starting to cool as it contacts the mould surfaces and spreads further from the point of entry.
- Uneven cooling during the filling process can lead to defects such as cold shuts (where two flow fronts meet and solidify too soon, preventing proper fusion). The hot core and cooler periphery can result in thermal gradients, which might affect the material properties after solidification.

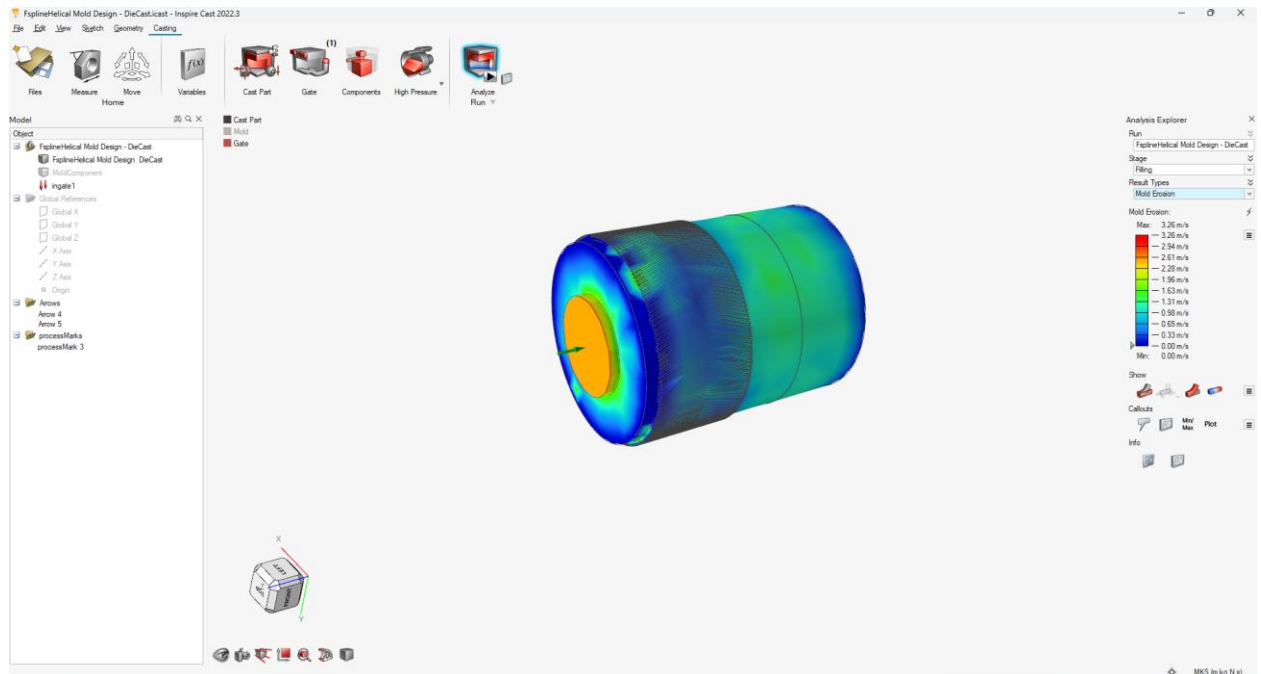


Figure 2.18: Mold erosion during filling

- **High Erosion Zones:** We can observe that the areas with yellow to red colours (near the inflow points) are subjected to the most wear, likely due to the high-speed molten material entering and impacting these regions.
- **Low Erosion Zones:** The blue and green regions indicate minimal erosion, likely due to lower velocities or less direct impact from the molten material.

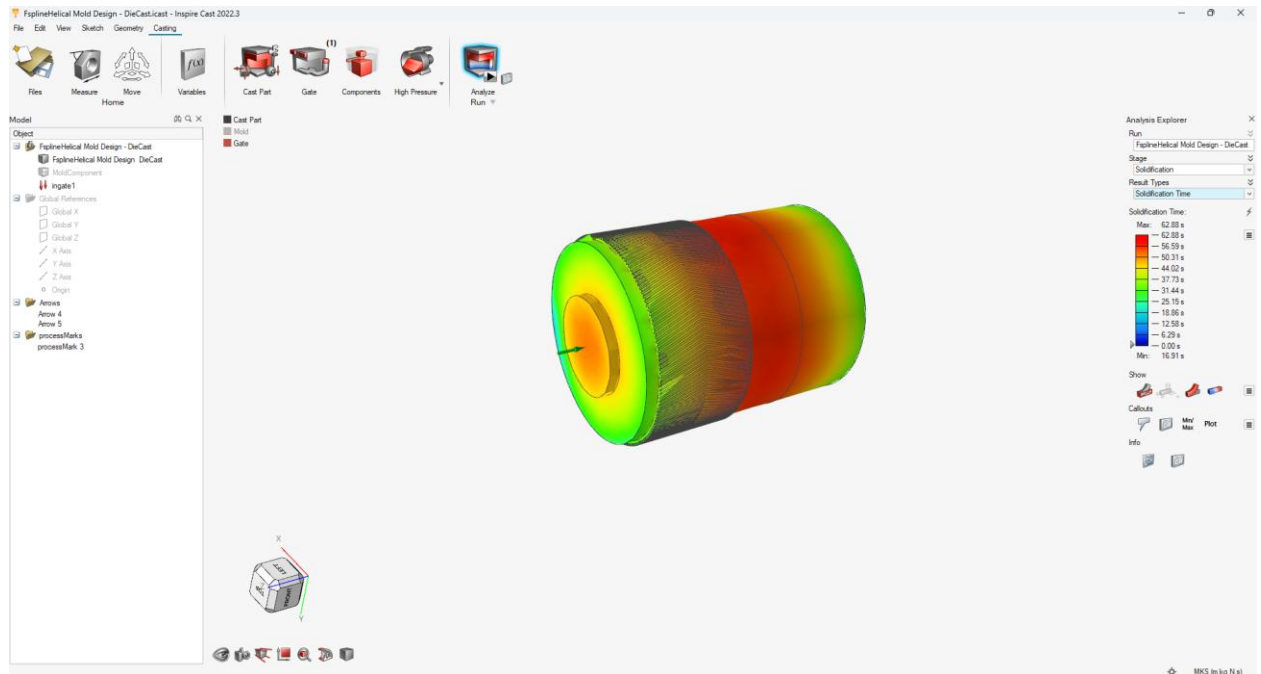


Figure 2.19: Variation of solidification time between the component

- **Red Zones (Long Solidification Time):** These areas tend to solidify last, which could be due to thicker sections or parts of the casting where the heat dissipation is slower. It could also indicate areas where hot spots are likely to form, increasing the risk of defects like porosity.
- **Green to Yellow Zones (Intermediate Solidification Time):** These zones represent areas with moderate solidification time, where cooling happens at a steady rate.
- **Blue Zones (Short Solidification Time):** These are areas of the cast part that solidify quickly, typically near the surface or thinner sections of the part. In this case, the outermost edge (closer to the mould walls) shows faster cooling.

2.7 DIE STRENGTH ANALYSIS (ANSYS FSI):

2.7.1 Conventional Strain Wave Gear:

Front Stroke:

Fluent Mesh Parameters:

- Method: Patch Conforming method
- Number of elements: 1648623

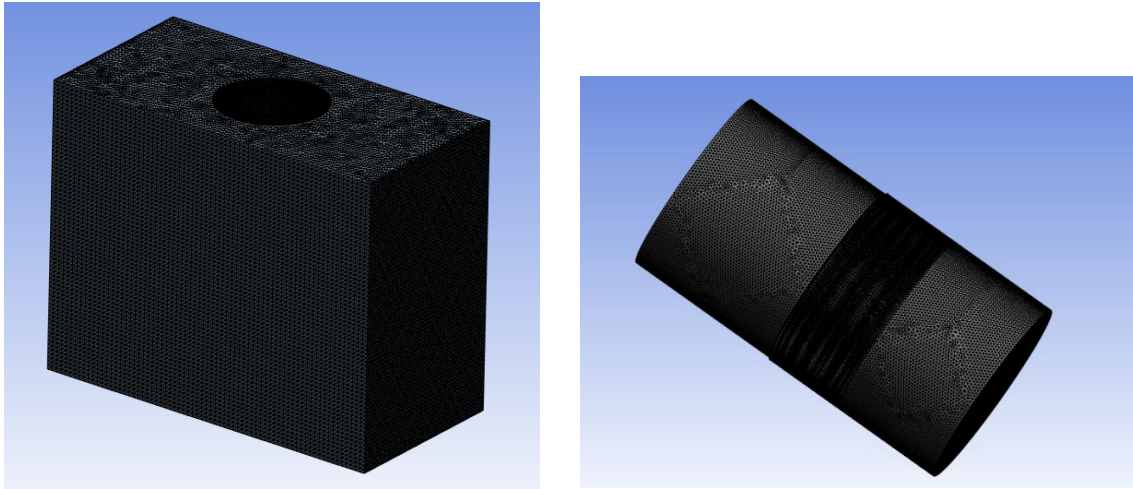


Figure 2.20

Fluent Mesh Quality:

	Minimum	Maximum	Average
Orthogonal quality	5.67e-008	0.993	0.685
Skewness	1.14e-006	1	0.313
Aspect Ratio	1.158	64283	2.86

Table 2.1

Fluent Parameters:

- Turbulence model: k-epsilon model
- Energy equation: ON
- Solving method: Coupled
- Boundary conditions: Pressure inlet – 80 MPa

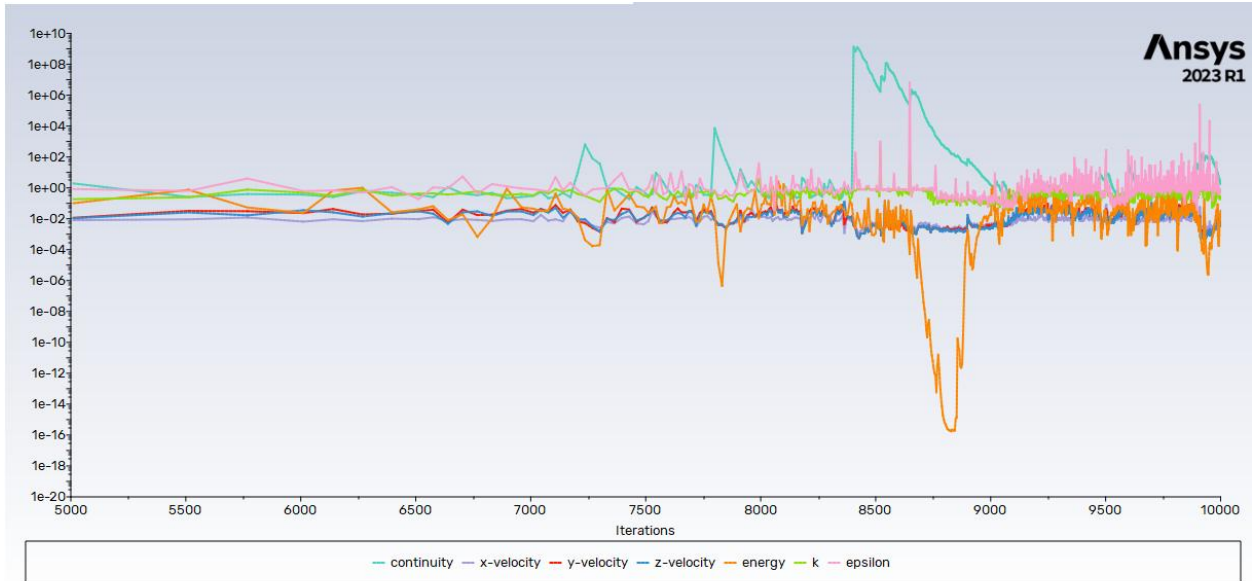


Figure 2.21: Residual Graph

Static Structural Parameters:

- **Imported pressure from Fluent**
- **Fixed support: All outer surfaces**

Static Structural Mesh Parameters:

- **Method: Patch Conforming method**
- **Number of elements: 223893**

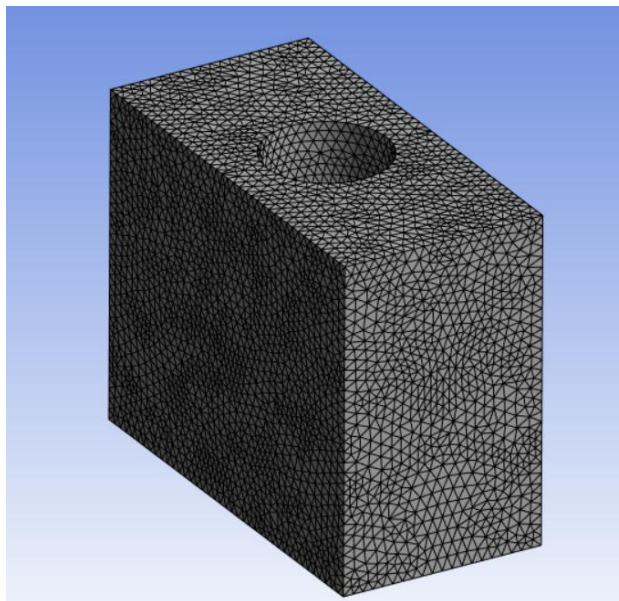


Figure 2.22

Static Structural Mesh Quality:

	Minimum	Maximum	Average
Element quality	2.52e-002	0.99	0.77
Skewness	4.43e-004	1	0.292
Aspect Ratio	1.165	220.81	3.144

Table 2.2

Simulation results:

Total Deformation:

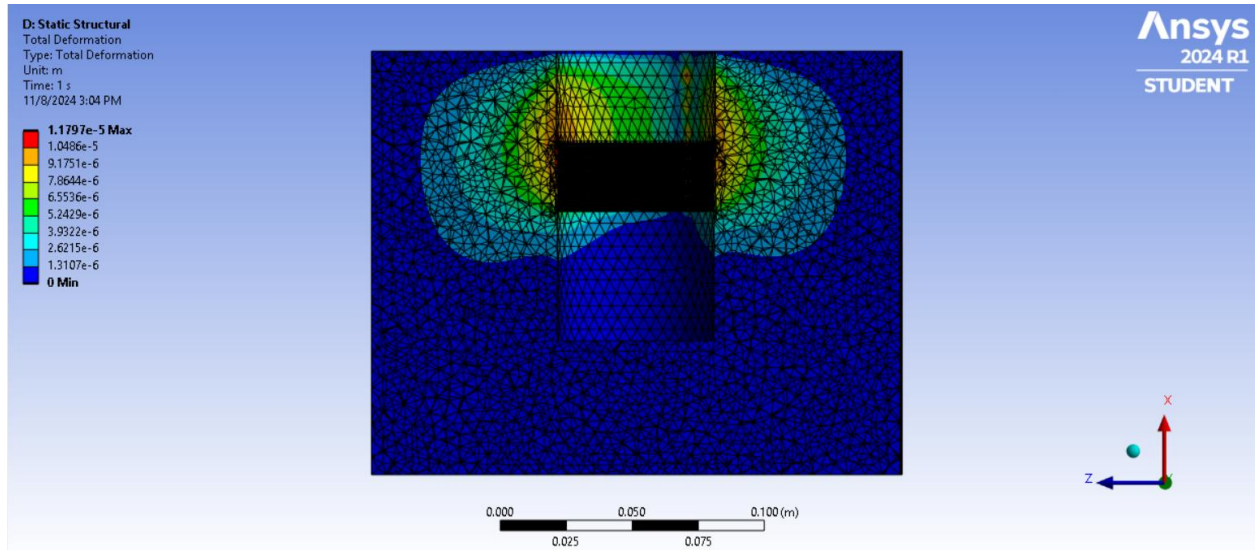


Figure 2.23

Equivalent (Von-Mises) Stress:

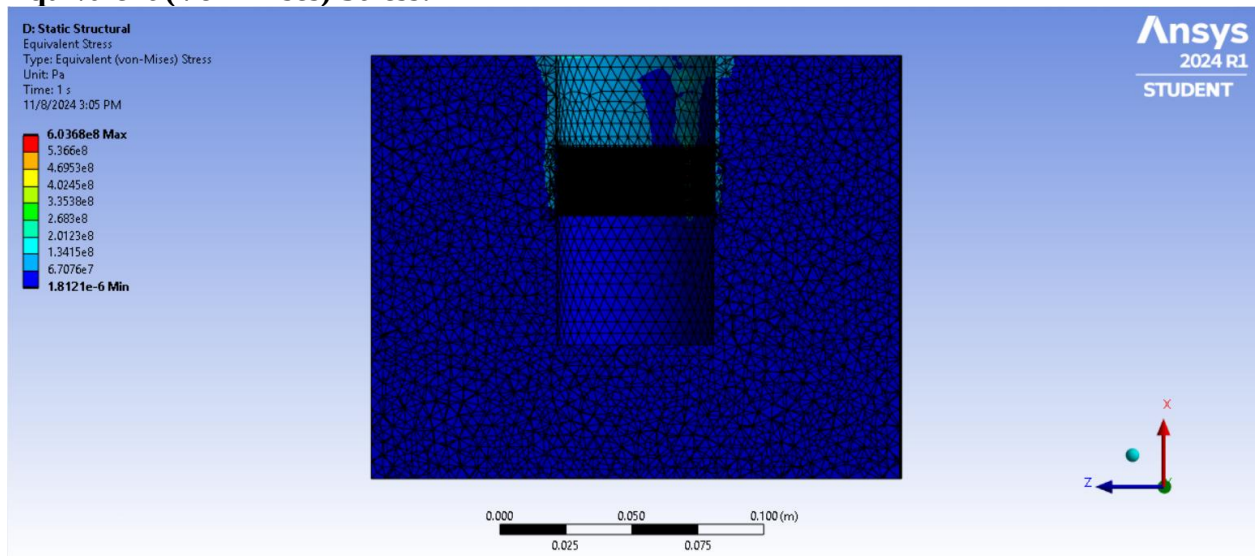


Figure 2.24

Equivalent (Von-Mises) Elastic Strain:

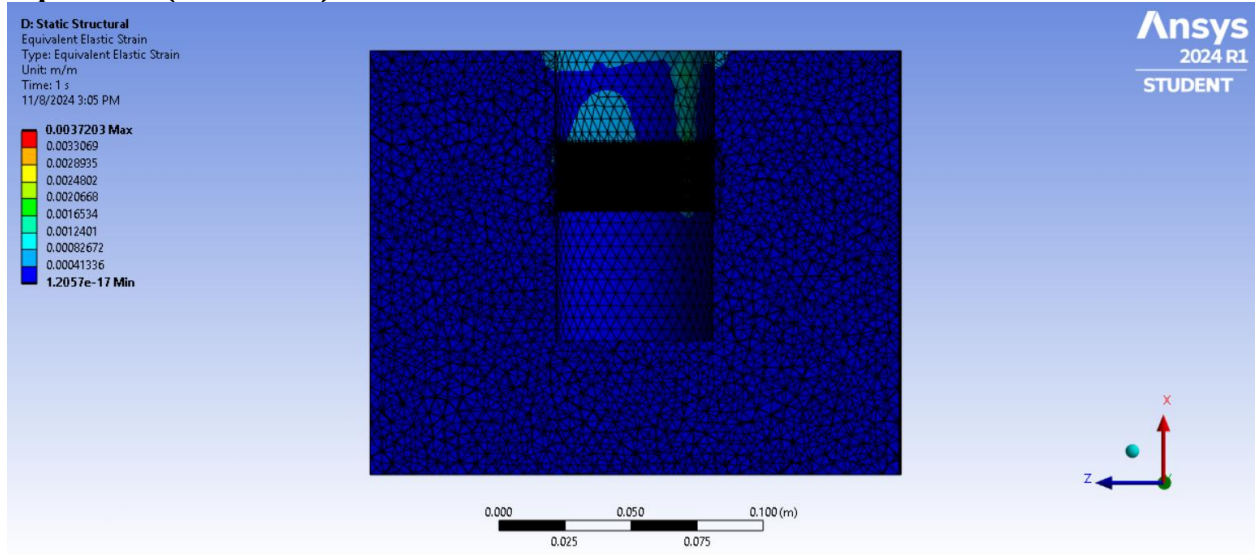


Figure 2.25

Back Stroke:

Fluent Mesh Parameters:

- Method: Patch Conforming method
- Number of elements: 2452352

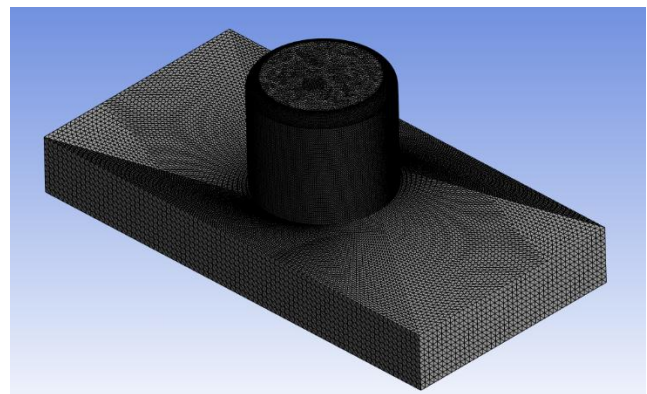
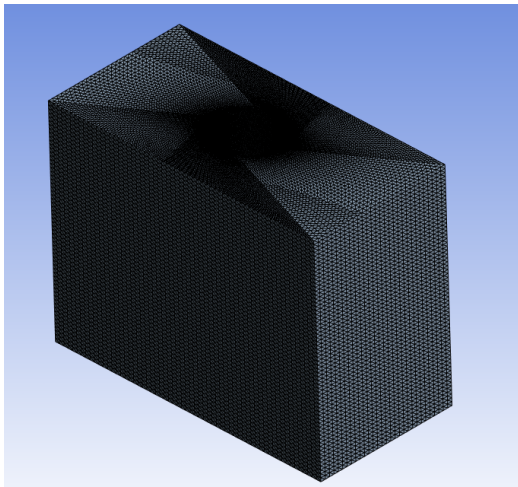


Figure 2.26

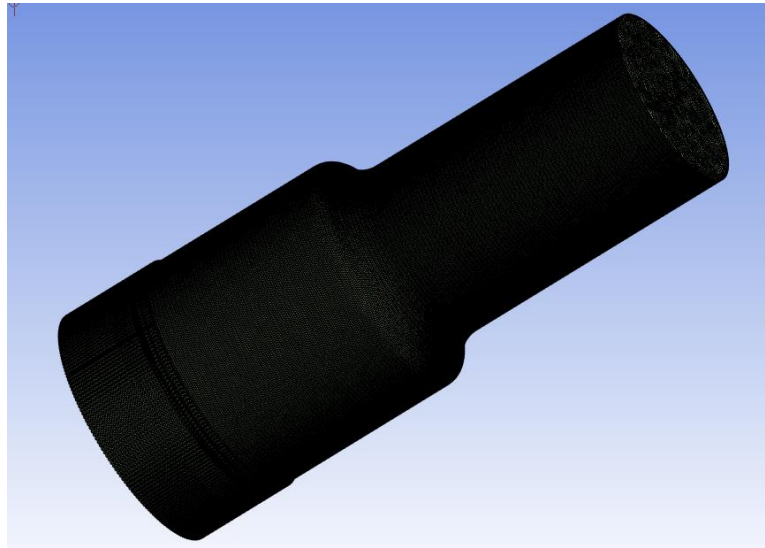


Figure 2.27: Fluid body

Fluent Mesh Quality:

	Minimum	Maximum	Average
Orthogonal quality	9.15e-007	0.99	0.65
Skewness	1.17	799.29	2.3
Aspect Ratio	6.35e-004	1	0.34

Table 2.3

Fluent Parameters:

- **Turbulence model: k-epsilon model**
- **Energy equation: ON**
- **Solving method: Coupled**
- **Boundary conditions: Pressure inlet – 80 MPa**

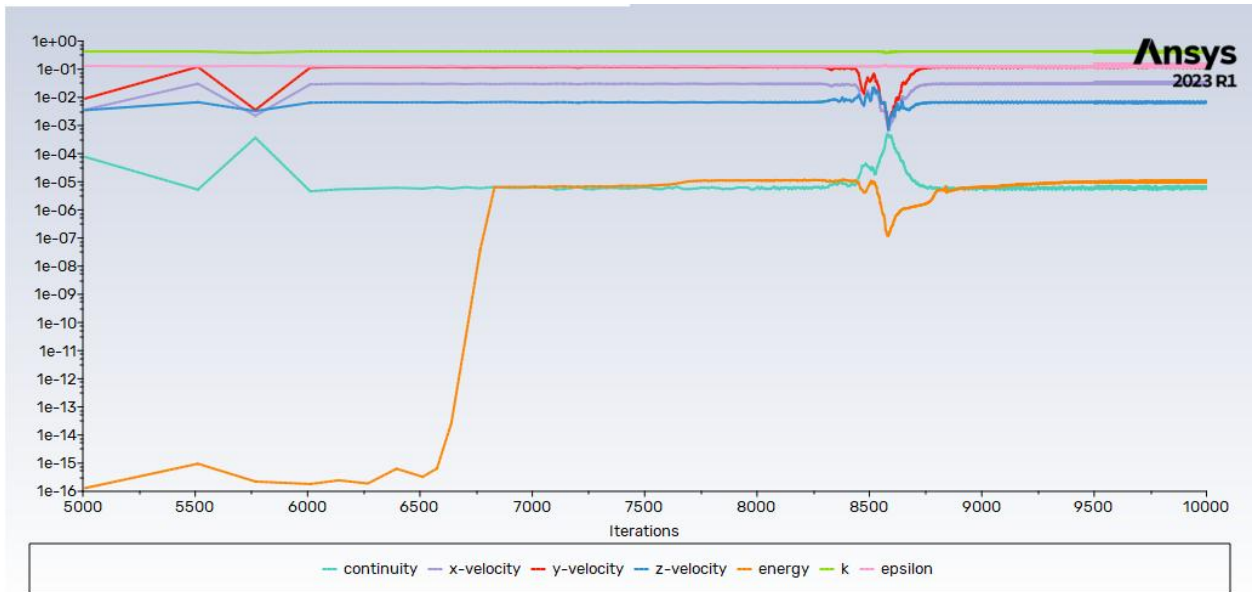


Figure 2.28: Residual Graph

Static Structural Parameters:

- Imported pressure from Fluent
- Fixed support: All outer surfaces

Static Structural Mesh Parameters:

- Method: Patch Conforming method
- Number of elements: 184336

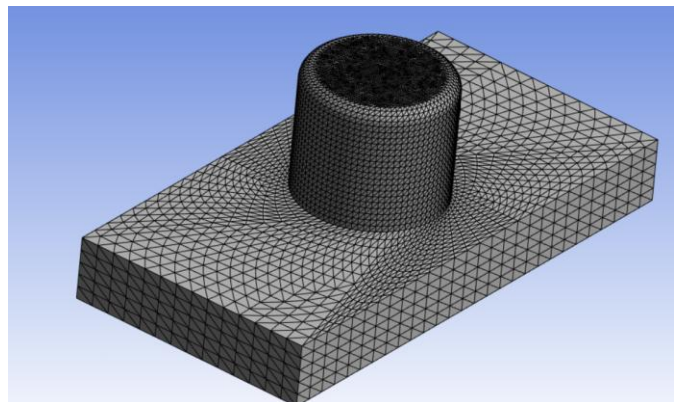
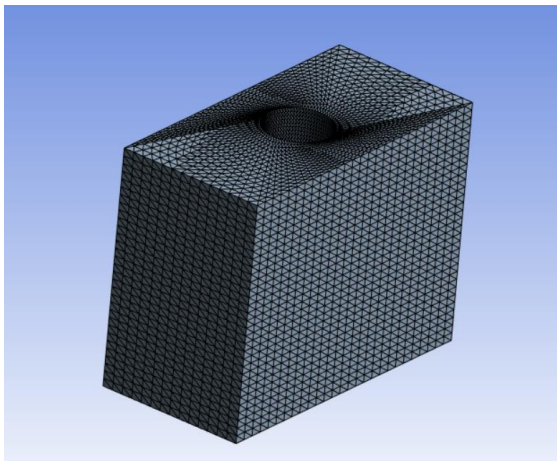


Figure 2.29

Static Structural Mesh Quality:

	Minimum	Maximum	Average
Element quality	8e-002	0.99	0.6
Skewness	5.15e-003	0.99	0.54
Aspect Ratio	1.19	39.07	2.96

Table 2.4

Simulation Results:

Total Deformation:

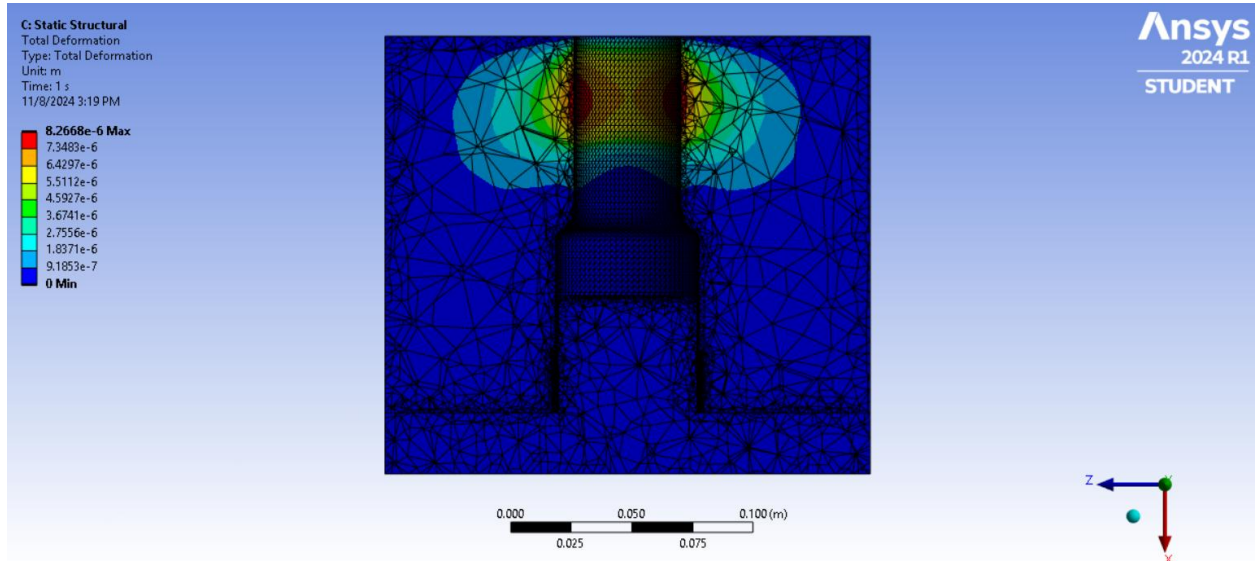


Figure 2.30

Equivalent (Von-Mises) Stress:

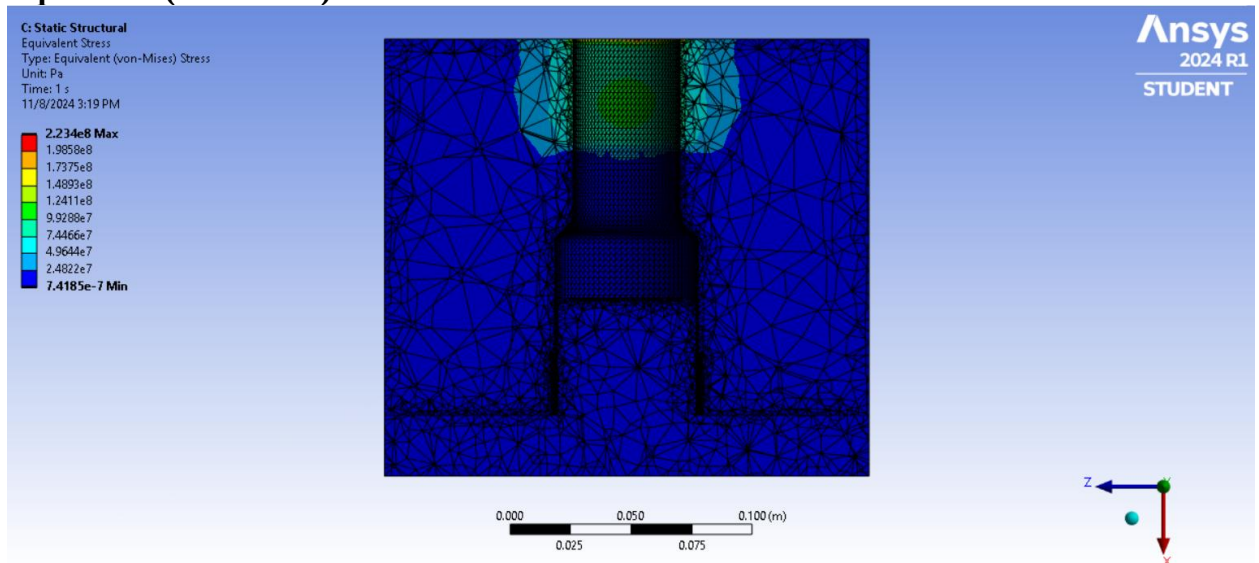


Figure 2.31

Equivalent (Von-Mises) Elastic Strain:

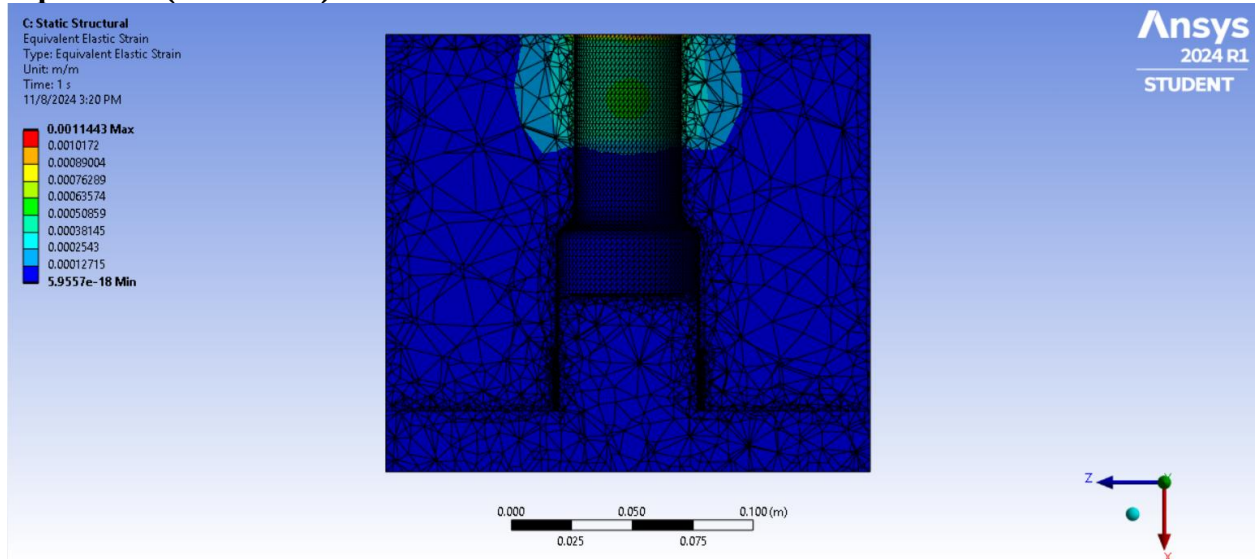


Figure 2.32

2.7.2 Conceptual Strain Wave Helical Gear:

Front Stroke:

Fluent Mesh Parameters:

- Method: Patch Conforming method
- Number of elements: 1955621

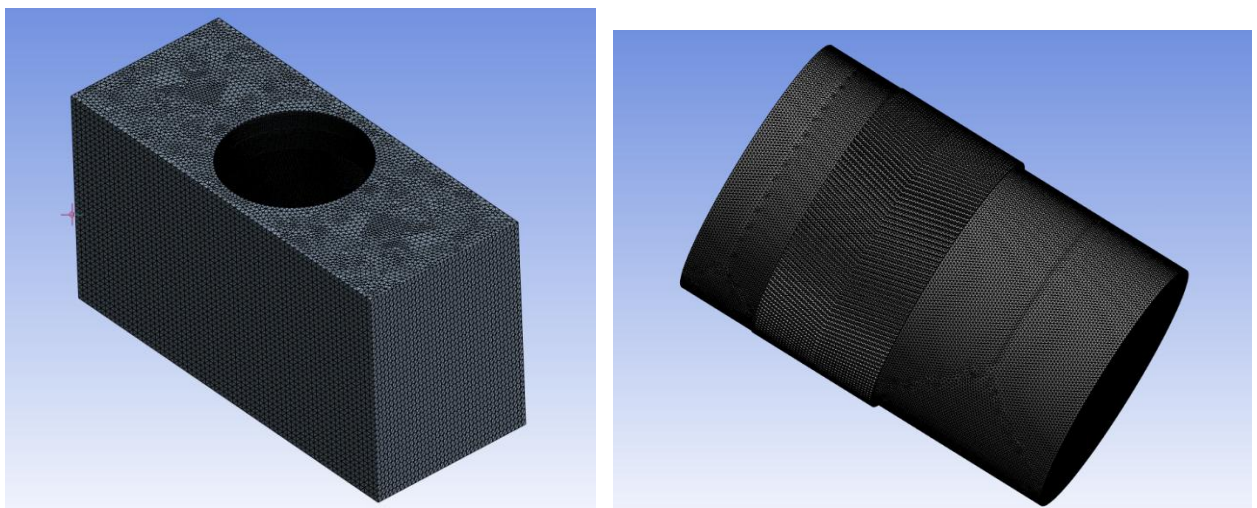


Figure 2.33

Fluent Mesh Quality:

	Minimum	Maximum	Average
Orthogonal quality	4.46e-002	0.99	0.73
Skewness	6.97e-005	0.95	0.26
Aspect Ratio	1.16	17.42	1.92

Table 2.5

Fluent Parameters:

- Turbulence model: k-epsilon model
- Energy equation: ON
- Solving method: Coupled
- Boundary conditions: Pressure inlet – 80 MPa

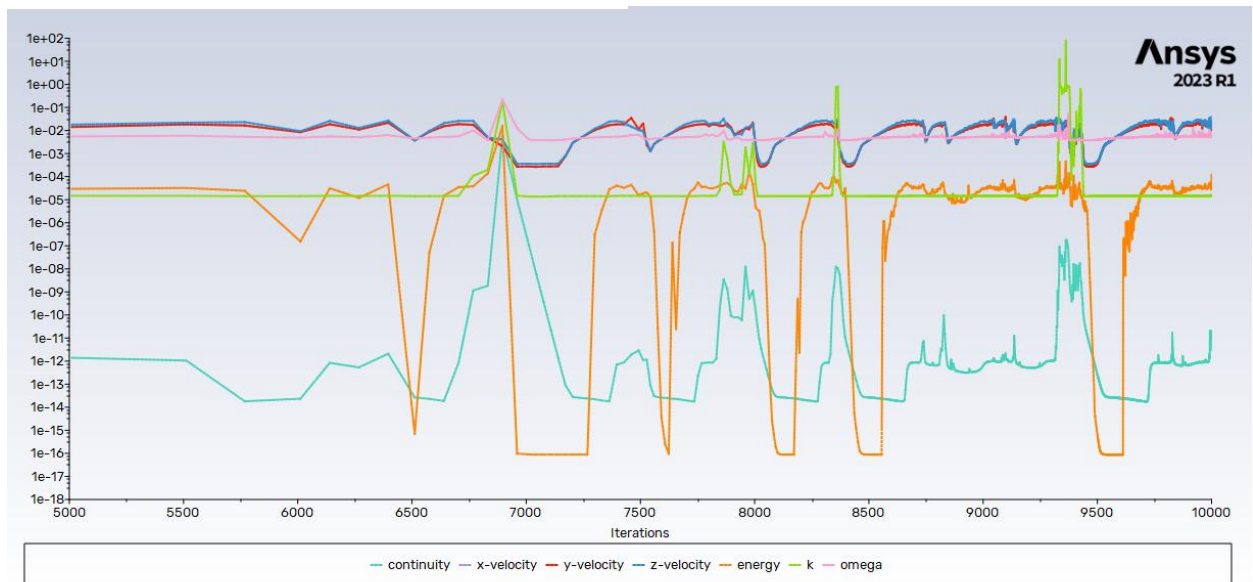


Figure 2.34: Residual Graph

Static Structural Parameters:

- Imported pressure from Fluent
- Fixed support: All outer surfaces

Static Structural Mesh Parameters:

- Method: Patch Conforming method
- Number of elements: 194163

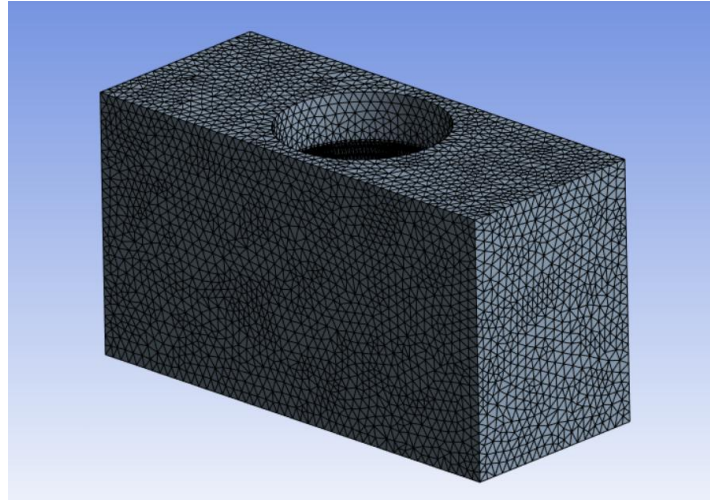


Figure 2.35: Mould component

Static Structural Mesh Quality:

	Minimum	Maximum	Average
Element quality	4.01e-002	0.99	0.76
Skewness	3.97e-004	0.99	0.3
Aspect Ratio	1.16	89.08	2.56

Table 2.6

Simulation results:

Total Deformation:

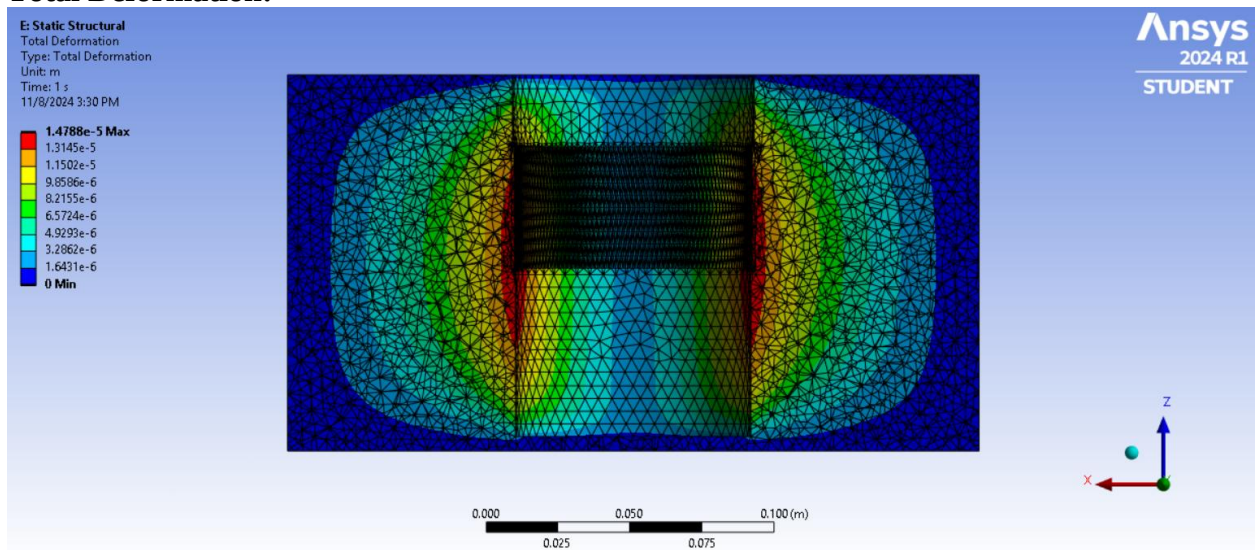


Figure 2.36

Equivalent (Von-Mises) Stress:

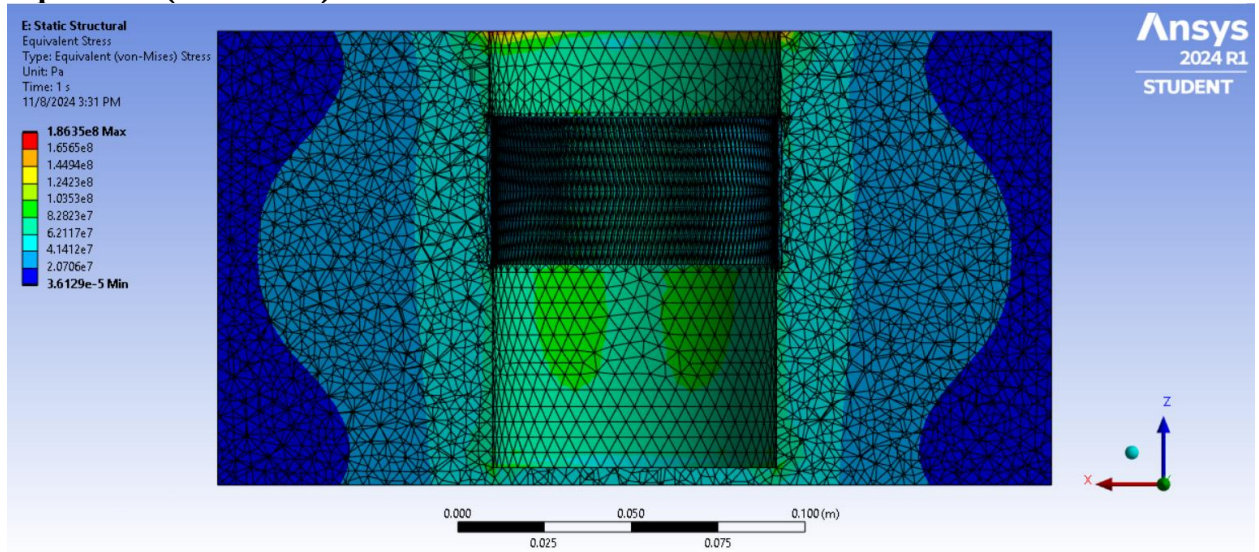


Figure 2.37

Equivalent (Von-Mises) Elastic Strain:

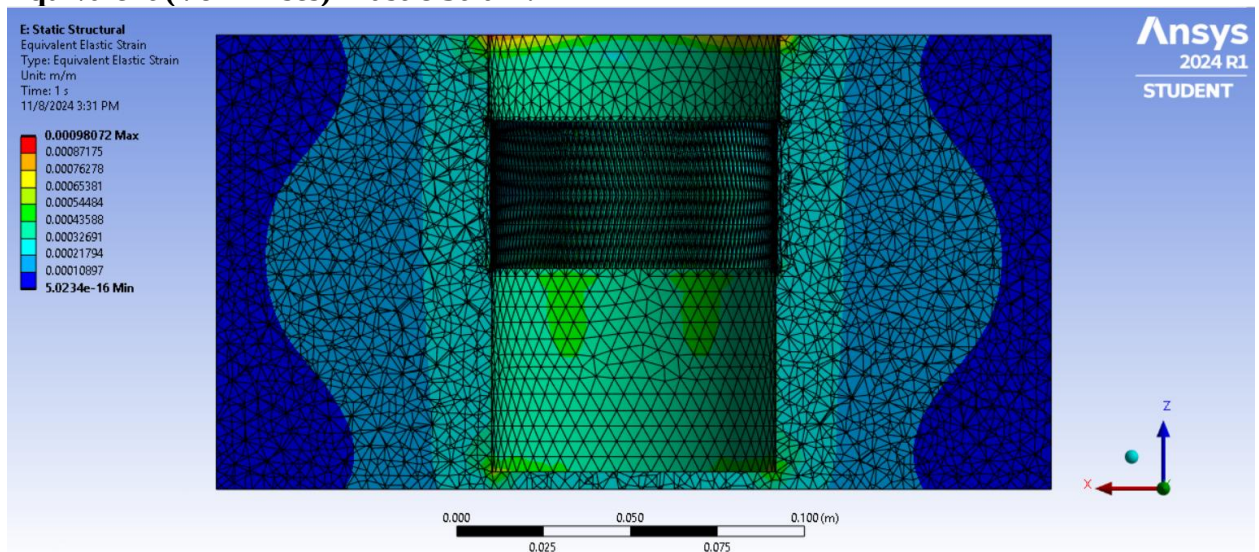


Figure 2.37

Back Stroke:

Fluent Mesh Parameters:

- **Method: Patch Conforming method**
- **Number of elements: 2460613**

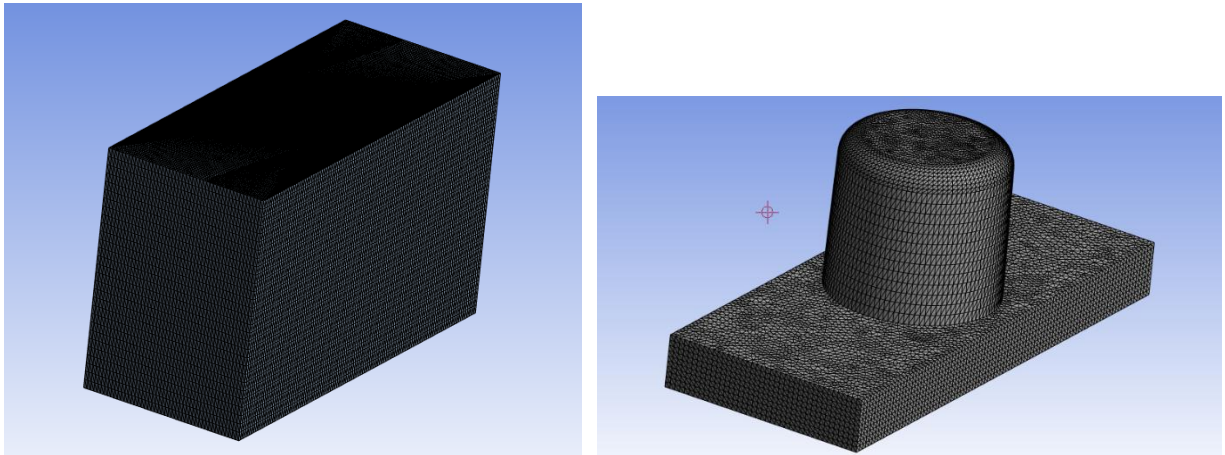


Figure 2.38: Mould Components

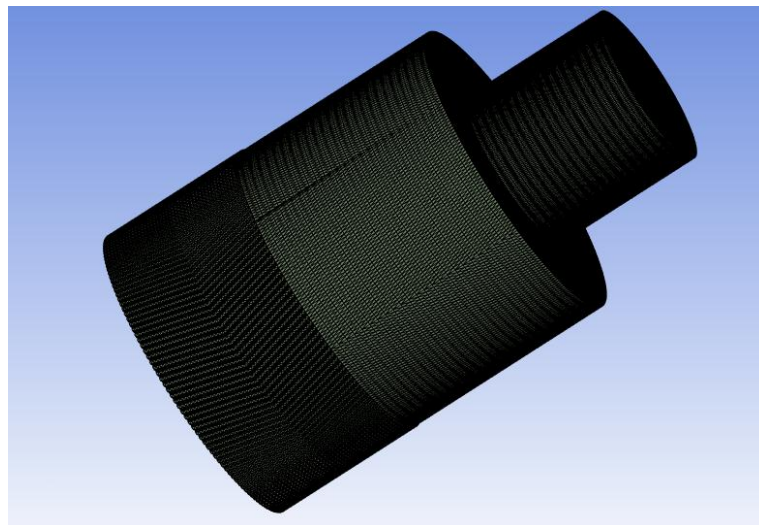


Figure 2.39: Fluid Body

Fluent Mesh Quality:

	Minimum	Maximum	Average
Orthogonal quality	4.72e-006	0.99	0.49
Skewness	4.75e-004	1	0.5
Aspect Ratio	1.16	3.26e+005	11.335

Table 2.7

Fluent Parameters:

- Turbulence model: k-epsilon model
- Energy equation: ON
- Solving method: Coupled
- Boundary conditions: Pressure inlet – 80 MPa

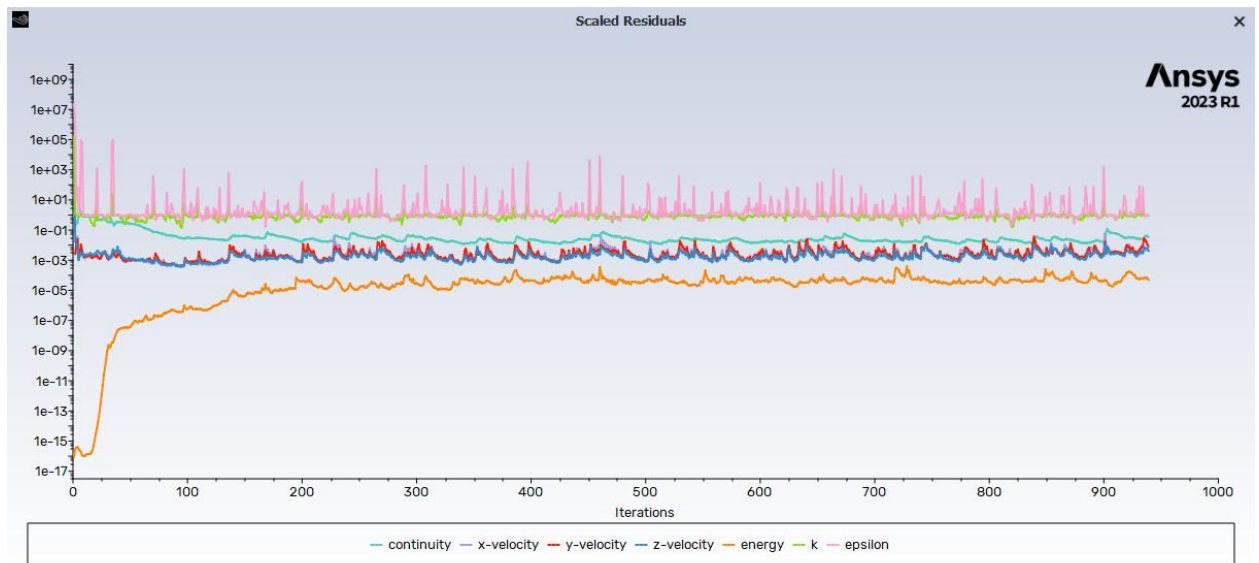


Figure 2.40: Residual Graph

Static Structural Parameters:

- Imported pressure from Fluent
- Fixed support: All outer surfaces

Static Structural Mesh Parameters:

- Method: Patch Conforming method
- Number of elements: 390626

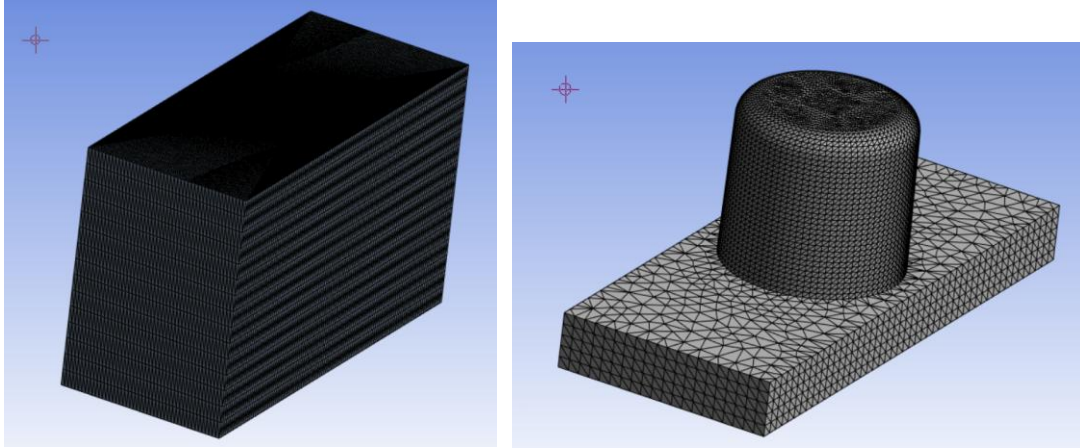


Figure 2.41: Mold components

Static Structural Mesh Quality:

	Minimum	Maximum	Average
Element quality	8e-002	0.99	0.6
Skewness	5.15e-003	0.99	0.54
Aspect Ratio	1.19	39.07	2.96

Table 2.8

Simulation Results:

Total Deformation:

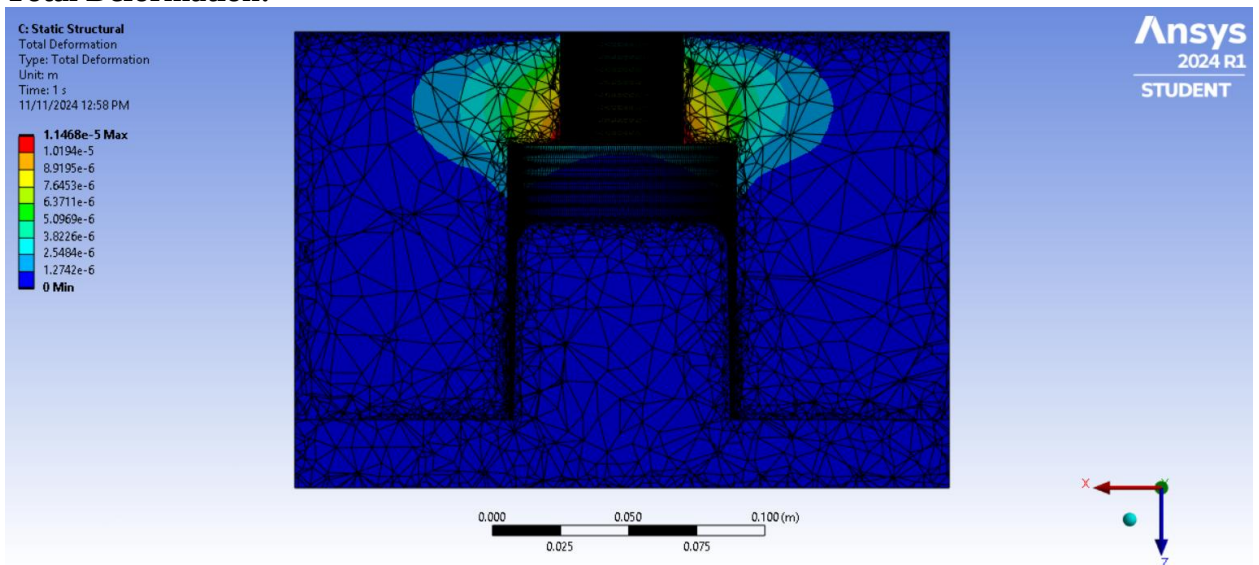


Figure 2.42

Equivalent (Von-Mises) Stress:

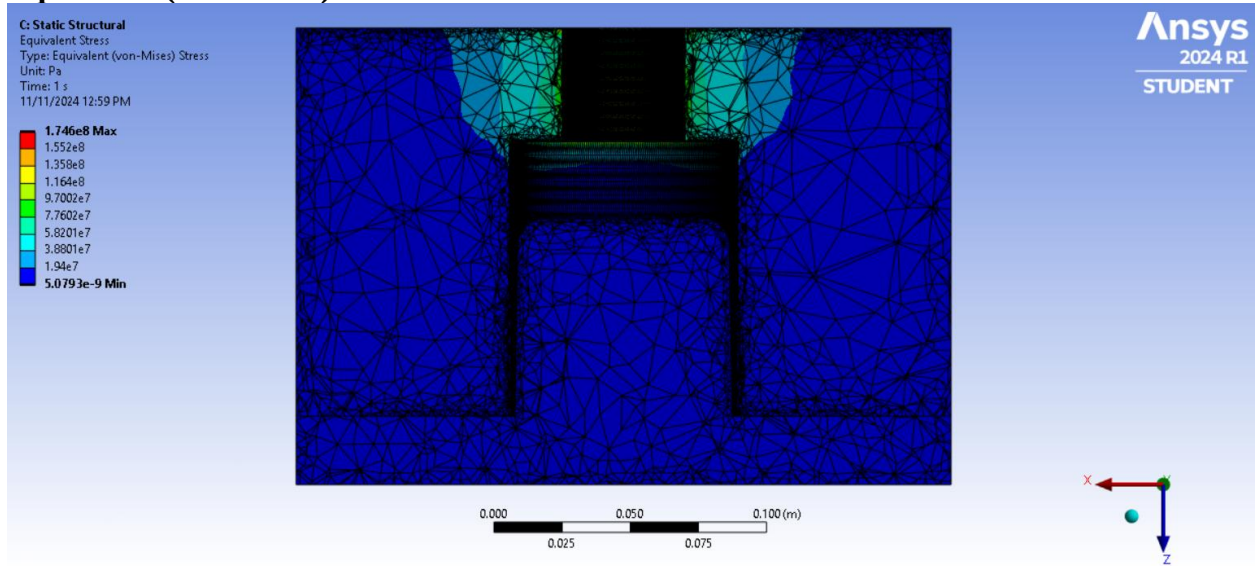


Figure 2.43

Equivalent (Von-Mises) Elastic Strain:

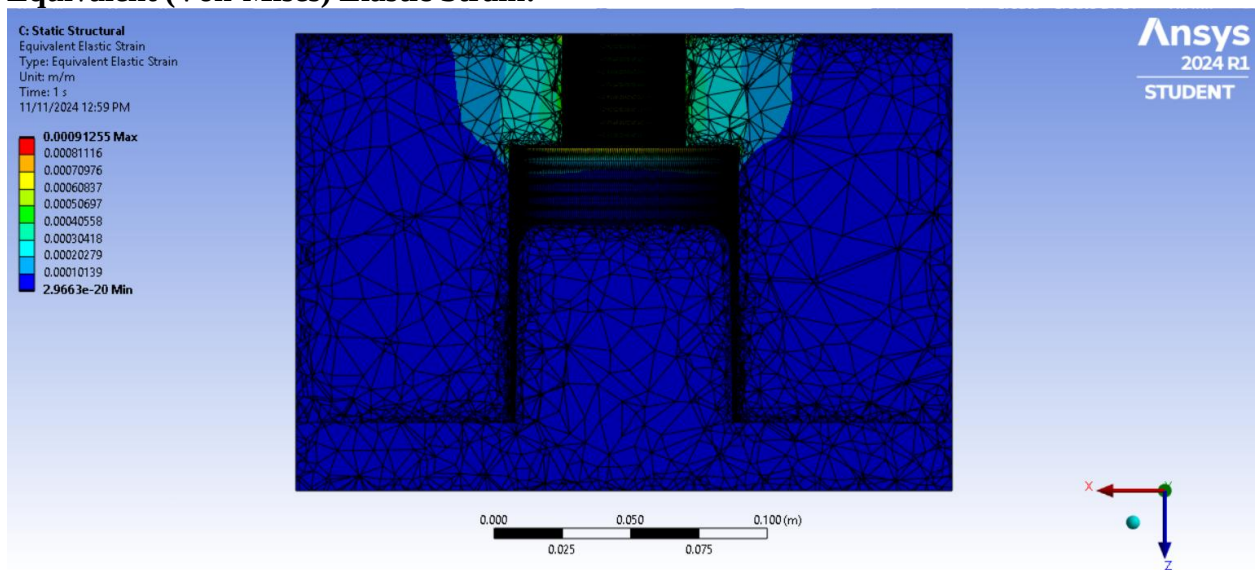


Figure 2.44

CHAPTER 3

RESULTS AND DISCUSSION

Benefits of using Strain wave helical gear over the conventional one in a harmonic drive:

1. **High Reduction Ratios:** Strain wave gears achieve high reduction ratios in a compact form, unlike conventional multi-stage systems.
2. **Zero Backlash:** Harmonic drives offer zero backlashes, critical for high-precision and repeatable positioning.
3. **High Torsional Stiffness:** They provide minimal deflection under load, ensuring accurate control in applications.
4. **High Torque Capacity:** The design allows high torque transmission with reduced localized wear and high load-bearing capacity.
5. **Smooth and Quiet Operation:** Strain wave gears operate smoothly and quietly, beneficial in low-noise environments.
6. **Long Service Life:** Distributed contact across teeth leads to lower wear and higher durability than conventional gears.
7. **Lightweight and Efficient:** These gears provide high performance in a lightweight design, ideal for mobile and energy-efficient applications.
8. **High Positioning Accuracy:** With zero-backlash and high stiffness, they deliver exceptional precision in controlled movements.

Stress Analysis: Conventional vs Conceptual Gear

Front Stroke:

Conventional	Conceptual
67 MPa	62.1 MPa

Table 3.1

Back Stroke:

Conventional	Conceptual
99.2 MPa	43.8 MPa

Table 3.2

Elastic Strain Analysis: Conventional vs Conceptual Gear

Front Stroke:

Conventional	Conceptual
0.000826 mm	0.000327 mm

Table 3.3

Back Stroke:

Conventional	Conceptual
0.000508 mm	0.000298 mm

Table 3.4

Deformation Analysis: Conventional vs Conceptual Gear

Front Stroke:

Conventional	Conceptual
6.53e-006	8.26e-006

Table 3.5

Back Stroke:

Conventional	Conceptual
5.51e-006	3.82e-006

Table 3.6

CHAPTER 4

CONCLUSION

This gear is used by DRDO in most of their components which require precise movement and high torque-carrying capacity.

The gears they are currently using offer good performance but with our gear we have proved that it can perform better when compared to the conventional one.

These gears are manufactured in both China and Germany as well, but the disadvantage is that the quality of the components produced in the Chinese Industry is not up to mark.

In Germany, these gears are of exceptional quality, but the cost is marginally higher.

So DRDO is looking into the Indian manufacturing industry to manufacture this gear at a good quality and lower cost, which we have proved in our project and conclude we will be able to manufacture our gears with good quality and lowered manufacturing cost.

Scope for Future Work:

As of now, we have only come up with an approximate result, but we will be coming up with an accurate result that can be used by the industry.

If the result is good, we are also planning on obtaining funds from VIT management to manufacture a mould and to prove that our project is experimentally correct as well.

If everything goes well, we will be publishing a research paper on this topic and applying for a patent for the mould design as well.

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Fabrication of Strain Wave Helical Gear

by Dean SMEC

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Heidar A. Talebi, Farzaneh Abdollahi, Rajni V. Patel, Khashayar Khorasani. "Neural Network-Based State Estimation of Nonlinear Systems", Springer Science and Business Media LLC, 2010

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Sustainable Development Goals related to the Project

SDG No	SDG Description	Yes/No
1	Eliminate Poverty	No
2	Erase Hunger	No
3	Establish Good Health and Well-Being	No
4	Provide Quality Education	No
5	Enforce Gender Equality	No
6	Improve Clean Water and Sanitation	No
7	Grow Affordable and Clean Energy	No
8	Create Decent Work and Economic Growth	No
9	Increase Industry, Innovation, and Infrastructure	Yes
10	Reduce Inequality	No
11	Mobilize Sustainable Cities and Communities	No
12	Influence Responsible Consumption and Production	No
13	Organize Climate Action	No
14	Develop Life Below Water	No
15	Advance Life On Land	No
16	Guarantee Peace, Justice, and Strong Institutions	No
17	Build Partnerships for the Goals	No

Justification

Strain wave gears are fundamental in precision applications, particularly in robotics, automation, and aerospace. Their unique design enables a high torque-to-weight ratio with minimal parts. This encourages advancements in compact, efficient systems, and fosters innovation. Strain wave gears can operate with high efficiency and minimal maintenance requirements, which reduces resource consumption over time. Their durability and efficiency make them suitable for industrial applications that emphasize resource conservation and low environmental impact, aligning with the goal of sustainable industrialization. By enabling high-precision control and reliable performance, strain wave gears are essential in infrastructure-sensitive applications such as robotics used in construction.

Technical / Manufacturing Readiness Level

Select the Domain / Sector (Choose one (or) more appropriate option from the dropdown)

Sl No	Domain Description	Yes/No
1	Agriculture & Rural Development innovations	No
2	Consumer Goods and Retail/Supply Chain/ Logistics	No
3	Defence & Security	No
4	Food Processing / Biotech	No
5	Healthcare & Biomedical devices	No
6	Cyber-physical systems, Blockchain, AI & ML Applications	No
7	Industry 4.0	No
8	IoT based technologies	No
9	Manufacturing (Additive Manufacturing, 3D printing, etc.)	Yes
10	Hydrogen and other alternative fuel technologies.	No
11	New / Innovative Metals and Materials	No
12	Other Emerging Areas Innovation for Start-up	No
13	Product Design, Development and Management	Yes
14	Renewable and Affordable Energy	No
15	Robotics and Drones	Yes
16	Smart Education	No
17	Smart Vehicles/ Electric vehicle/ Electric vehicle motor and battery technology	No
18	Sustainable Environment	No
19	Waste Management/Waste to Wealth Creation	No
20	Circular economy and zero waste technologies	No

Justification

Strain wave Helical gear is renowned for high precision and achieves smooth, efficient, and backlash-free movement. Their capability to operate with minimal backlash significantly improves the stability and accuracy of robotic movements. Casting techniques are used in manufacturing durable components while Electrochemical Machining (ECM) can be applied to refine intricate parts. The development of strain wave gears incorporates advanced materials and optimized gear profiles, allowing manufacturers to meet stringent application-specific requirements. Product management benefits as strain wave gears are modular and standardized to some extent, easing integration across multiple product lines and supporting lifecycle management through predictable performance metrics and reliability standards.

Select the Innovation Type

SI No	Innovation Type	Yes/No
1	Product	Yes
2	Process	Yes
3	Service	No
4	Market Place	No
5	Business/Management	No

Justification

Strain wave gears are designed with a unique structure involving a flexible spline that deforms to match an outer rigid spline, enabling exceptionally high torque transmission with minimal backlash and a compact form factor. The innovation lies in the efficiency and compactness of the gear system, enabling products to be smaller, lighter, and more accurate. These characteristics are key to modern high-precision fields, making strain wave gears a valuable product innovation in engineering. Innovations in the manufacturing process have improved the efficiency and scalability of producing strain wave gears. HPDC and ECM techniques reduce production time, cost, and waste while ensuring high quality and precision. As a result, manufacturers can offer these high-performance gears to a broader market and integrate them into various applications more effectively.

Select the Technical Readiness Level (Choose one (or) more appropriate option from the dropdown)

SI No	Readiness Level	Yes/No
1	Basic research, Principles postulated observed but no experimental proof available	Yes
2	Technology formulation. The concept and application have been formulated	No
3	Applied research. First laboratory tests completed; proof of concept	No
4	Small-scale prototype built in a laboratory environment (ugly prototype)	No
5	Large-scale prototype tested in the intended environment	No
6	Prototype system tested in an intended environment close to expected performance	No
7	Demonstration system operating in the operational environment at a pre-commercial scale	No
8	First-of-a-kind commercial system. Manufacturing issues solved	No
9	Full commercial application, the technology available for consumers	No

Justification

Strain wave gears operate on the concept of elastic deformation to transmit torque, allowing for precise, high-ratio reductions with minimal backlash. The theoretical mechanics of these principles are postulated in engineering literature and foundational studies, explaining the deformation wave and mechanical interactions within the gear. The operational principles of strain wave gears have been analyzed theoretically, and mathematical models exist to describe their potential performance. However, these models remain largely untested in practical scenarios, with assumptions about material behaviour, stress distributions, and dynamic loading that are not yet verified experimentally. Practical evidence will be published to validate the complete theoretical models once the fabrication is done.

Select the Manufacturing Readiness Level (Choose one (or) more appropriate option from the dropdown)

SI No	Readiness Level	Yes/No
1	Basic manufacturing implications identified	Yes
2	Manufacturing concepts identified	Yes
3	Manufacturing proof of concept developed	No
4	Capability to produce the technology in a laboratory environment	No
5	Capability to produce prototype components in a production-relevant environment	No
6	Capability to produce a prototype system or subsystem in a production-relevant environment	No
7	Capability to produce systems, subsystems, or components in a production-representative environment	No
8	Pilot line capability demonstrated. Ready to begin low-rate production.	No
9	Low rate production demonstrated. Capability in place to begin Full Rate Production.	No
10	Full rate production demonstrated and lean production practices in place	No

Justification

Manufacturing processes must ensure material integrity under cyclical loading, necessitating strict adherence to quality and material standards. The Strain wave gears involve intricate components, which require high precision in their internal and external tooth profiles to maintain the deformation and load-transmission characteristics so advanced machining techniques such as wire ECM (Electrochemical Machining) are used. CAM (Computer Aided Manufacturing) software facilitates the high-precision machining required for strain wave gears, optimizing tool paths, speeds, and feeds to achieve the desired geometric accuracy. CAM enables rapid prototyping of strain wave gears, allowing for iterative design improvements with minimal time lost between iterations and enhancing manufacturability assessments.

Course Outcome

CO No	Course Outcome	Rating
1	Formulate specific problem statements for ill-defined real-life problems with reasonable assumptions and constraints	5
2	Perform a literature search and/or patent search in the area of interest	5
3	Develop a suitable solution methodology for the problem	5
4	Conduct experiments / Design & Analysis/solution iterations and document the results	5
5	Perform error analysis/benchmarking / costing	5
6	Synthesise the results and arrive at scientific conclusions/products / solution	5
7	Document the results in the form of technical report/presentation	5

Program Outcome

SI No	Programme Outcome	Yes/No
1	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.	Yes>
2	Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.	Yes>
3	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations	Yes>
4	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions for complex problems	Yes>
5	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.	Yes>
6	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.	Yes>
7	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.	Yes>
8	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.	Yes>
9	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings	Yes>
10	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	Yes>
11	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.	Yes>
12	Recognize the need for and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.	Yes>